Statistics

Chapter 10 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 69

PROBLEM STATEMENT

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Solution A

Step 1

a.) Recall that one-tailed test should be use when the hypothesis has an inequality sign and two-tailed test when the hypothesis has an equality sign. Since the given hypothesis has a sign of equality, then it should use

$$\text{two-tailed test}$$

where the rejection region is on the left side and the right side of the curve because the alternate hypothesis has an inequality sign.

Step 2

b.) To determine the decision rule, it needed to find the critical value. Since the rejection region is on the both side of the curve with a significance level α of 0.05, then the standard normal distribution will be

$$P(z > z_0 \text{ or } z_0 < z) = \frac{0.05}{2} = 0.025.$$

Step 3

Find the area 0.025 in the z-score table to obtain the critical value. By looking at the table:

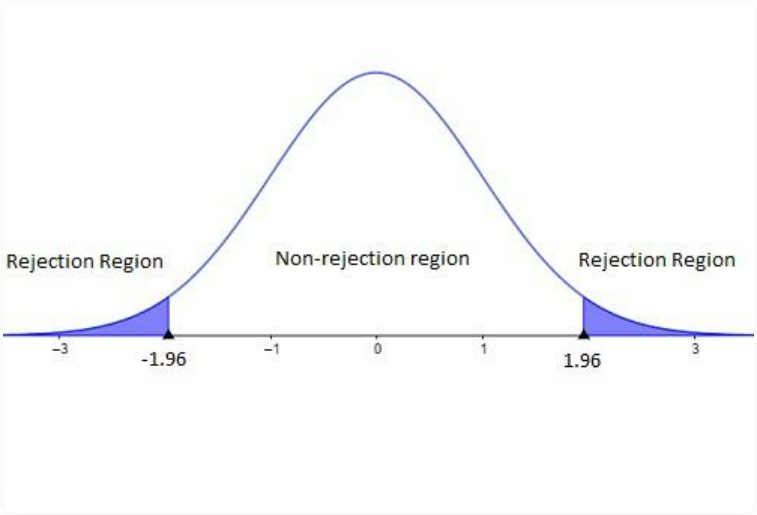
z	$\cdots$	0.06	$\cdots$
$\vdots$	$\ddots$	$\vdots$	
-1.6	$\cdots$	0.025	
$\vdots$			

the area will be found in the row head of -1.6 and the column head of 0.06. Taking the sum of absolute value of -1.9 and 0.06, thus

$$|-1.9| + 0.06 = 1.96$$

Step 4

To visualize where the critical value is, the graph is represented below.



The rejection region is on the both tail of the curve because the alternative hypothesis has an inequality sign. Which means that if the test statistics is greater than the critical value 1.96 or less than the critical value -1.96, the null hypothesis will be rejected.

Step 5

Since the rejection region is in the both tail of the curve, then fail to reject the null hypothesis if the test statistic is less than or equal to 1.96 and greater than or equal to -1.96. The decision rule will be

$$\text{When z} > 1.96 \text{ or z} < \text{-}1.96, \text{ reject } H_0$$

Step 6

c.) The test statistics that will be using is a one-sample test

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}},$$

where:

- z is the test statistics,
- $\bar{x}$ is the sample mean,
- μ is the population mean,
- σ is the population standard deviation, and
- n is the sample size.

The one-sample test should be used in the given claim because it has a one-parameter comparing to the given population mean assuming that it's normally distributed.

Step 7

By substituting the sample size n is 36, sample mean $\bar{x}$ of 49, population mean μ of 50, and population standard deviation σ of 5 into the formula for the test statistic of two-tailed test, thus

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{49 - 50}{5/\sqrt{36}} = -1.2.$$

In conclusion, the value of the test statistic is

$$z = -1.2$$

Step 8

d.) Since the test statistic is -1.2 which is greater than the critical value -1.96 but less than the critical value 1.96, then the null hypothesis is fail to be rejected. The decision will be

- At 0.05 significance level **fail** to reject H_0.

Step 9

e.) The p-value or probability value represents the probability of how likely it is to obtain the value of the test statistics given that the null hypothesis is true. To find the p-value, assume that H_0 is true. Then the standard normal distribution will be

$$P(z < -1.2)$$

Step 10

Multiply the standard normal distribution by 2 since it's a two-tailed and find the area in the z-score table, thus

$$p = 2P(z < -1.2)$$

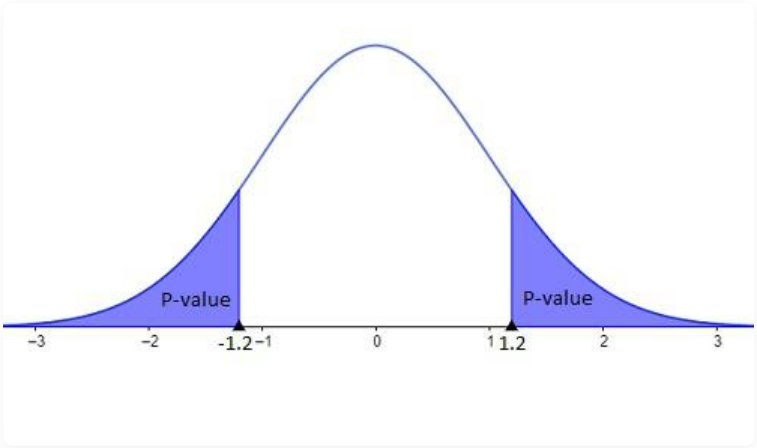
Using the z-score table to find the area of the probability $P(z < -1.2)$, by looking up at the table:

z	$\cdots$	0.00	$\cdots$
$\vdots$	$\ddots$	$\vdots$	
-1.2	$\cdots$	0.11507	
$\vdots$			

Substitute the area of $P(z < -1.2)$ found in the z-table into the equation, thus

$$p = 2(0.11507) = 0.23014 \approx 0.2302.$$

To visualize the p-value, the graph is represented below where the p-value is the shaded region.



Step 11

In conclusion the, the p-value is

$$0.2302$$

which is having an 23.02% chance to find a z value equal to or greater than to it.

Result

$a.)$ Two-tailed test

$b.)$ When z < -1.2 or z > 1.2, reject H_0

$c.)$ $z = -1.2$

$d.)$ fail to reject H_0

$e.)$ $p = 0.2302$

Solution B

Step 1

Given:

$$H_0 : \mu = 50$$
$$H_a : \mu \neq 50$$
$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 36$$
$$\overline{x} = \text{Sample mean} = 49$$
$$\sigma = \text{Population standard deviation} = 5$$

(a) If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Two-tailed test

Step 2

(b) The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right and all z-scores below the z-score $-z_0$ that has a probability of 0.05 to its left.

$$P(z < -z_0 \text{ or } z > z_0) = 0.05$$

Since the standard normal distribution is symmetric about 0:

$$P(z < -z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of 0.025 in the normal probability table of the appendix, while z_0 is this z-score with an opposite sign.

$$z_0 = 1.96$$

Thus the rejection region then contains all z-scores smaller than -1.96 and all z-scores greater than 1.96, which implies that the decision rule is: Reject the null hypothesis when the z-score is not between -1.96 and 1.96.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean from the null hypothesis, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{49 - 50}{5/\sqrt{36}} \approx -1.20$$

Step 4

(d) Z-score found in part (c): $z = -1.20$

Decision rule found in part (b): Reject the null hypothesis when the z-score is not between -1.96 and 1.96.

We note that the z-score of -1.20 lies between -1.96 and 1.96, which implies that we fail to reject the null hypothesis H_0.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.20$ or larger than $z = 1.20$.

$$P = P(z < -1.20 \text{ or } z > 1.20)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.20)$ is given in the row starting with "1.2" and in the column starting with ".00" of the table.

$$\begin{aligned} P &= P(z < -1.20 \text{ or } z > 1.20) \\ &= 1 - P(-1.20 < z < 1.20) \\ &= 1 - 2P(0 < z < 1.20) \\ &= 1 - 2(0.3849) \\ &= 0.2302 \end{aligned}$$

Result

(a) Two-tailed test

(b) Reject the null hypothesis when the z-score is not between -1.96 and 1.96.

(c) $z = -1.20$

(d) Fail to reject the null hypothesis H_0.

(e) $P = 0.2302$

PROBLEM STATEMENT

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Step 1

Given:

$$\begin{aligned} H_0 &: \mu \leq 10 \\ H_1 &: \mu > 10 \\ \alpha &= \text{Significance level} = 0.02 \\ n &= \text{Sample size} = 36 \\ \overline{x} &= \text{Sample mean} = 12 \\ \sigma &= \text{Population standard deviation} = 3 \end{aligned}$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

Step 2

(b) The rejection region of a right-tailed test with $\alpha = 0.02$ contains all z-scores above the z-score z_0 that has a probability of 0.02 to its right.

$$P(z > z_0) = 0.02$$

Use the **Complement rule**: $P(A^c) = P(\text{ not } A) = 1 - P(A)$

$$P(z < z_0) = 1 - P(z > z_0) = 1 - 0.02 = 0.98$$

Let us determine the z-score that corresponds with a probability of $0.98 - 0.5 = 0.48$ in the normal probability table of the appendix. We note that the probability closest to 0.48 is 0.4798, which is given in the row starting with "2.0" and in the column starting with ".05", thus the corresponding z-score is $2.0 + .05 = 2.05$.

$$z_0 = 2.05$$

Thus the rejection region then contains all z-scores greater than 2.05.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{12 - 10}{3/\sqrt{36}} \approx 4.00$$

Step 4

(d) Result part (b): $z > 2.05$ results in the rejection of the null hypothesis H_0

Result part (c): $z = 4.00$

Since the z-score of 4.00 exceeds 2.05, we reject the null hypothesis H_0

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is right-tailed, the P-value is the probability that the z-score is larger than $z = 4.00$.

$$P = P(z > 4.00)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 4.00)$ is given in the row starting with "4.0" and in the column starting with ".00" of the table.

$$\begin{aligned} P &= P(z > 4.00) \\ &= 1 - P(z < 4.00) \\ &= 1 - P(z < 0) - P(0 < z < 4.00) \\ &\approx 1 - 0.5000 - 0.5000 \\ &= 0 \end{aligned}$$

🎯 **Result**

(a) Right-tailed

(b) $z > 2.05$

(c) $z = 4.00$

(d) Reject H_0

(e) About 0

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned} H_0 &: \mu \leq 20 \\ H_1 &: \mu > 20 \\ \alpha &= \text{Significance level} = 0.05 \\ n &= \text{Sample size} = 36 \\ \overline{x} &= \text{Sample mean} = 21 \\ \sigma &= \text{Population standard deviation} = 5 \end{aligned}$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

Step 2

(b) The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Use the **Complement rule**: $P(A^c) = P(\text{ not } A) = 1 - P(A)$

$$P(z < z_0) = 1 - P(z > z_0) = 1 - 0.05 = 0.95$$

Let us determine the z-score that corresponds with a probability of $0.95 - 0.5 = 0.45$ in the normal probability table of the appendix. We note that the probability closest to 0.45 is 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{21 - 20}{5/\sqrt{36}} \approx 1.20$$

Step 4

(d) Result part (b): $z > 1.65$ results in the rejection of the null hypothesis H_0

Result part (c): $z = 1.20$

Since the z-score of 1.20 does not exceed 1.65, we fail to reject the null hypothesis H_0.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is right-tailed, the P-value is the probability that the z-score is larger than $z = 1.20$.

$$P = P(z > 1.20)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.20)$ is given in the row starting with "1.2" and in the column starting with ".00" of the table.

$$\begin{aligned} P &= P(z > 1.20) \\ &= 1 - P(z < 1.20) \\ &= 1 - P(z < 0) - P(0 < z < 1.20) \\ &= 1 - 0.5000 - 0.3849 \\ &= 0.1151 \\ &= 11.51\% \end{aligned}$$

Result

(a) Right-tailed

(b) $z > 1.65$

(c) $z = 1.20$

(d) Fail to reject H_0

(e) $0.1151 = 11.51\%$

Solution B

Step 1

a.) Recall that one-tailed test should be used when the hypothesis has an inequality sign and two-tailed test when the hypothesis has an equality sign. Since the given hypothesis has a sign of inequality, then it should use

$$\text{one-tailed test}$$

where the rejection region is on the right side of the curve because the alternate hypothesis has an inequality sign of greater than.

Step 2

b.) To determine the decision rule, it needed to find the critical value. The critical value is the threshold whether the null hypothesis should be rejected or fail to be rejected. Based on the alternative hypothesis, the rejection region is on the right tail of the curve with a significance level α of 0.05, then the standard normal distribution will be

$$P(z > z_0) = 0.05$$

Step 3

Apply the complement rule to the standard normal distribution to find the area of the non-rejection region. Thus,

$$1 - P(z > z_0) = 1 - 0.05 = 0.95$$

Find the area 0.95 in the z-score table to find the critical value. Then take the sum of the row and the column where the area is placed. By looking at the table:

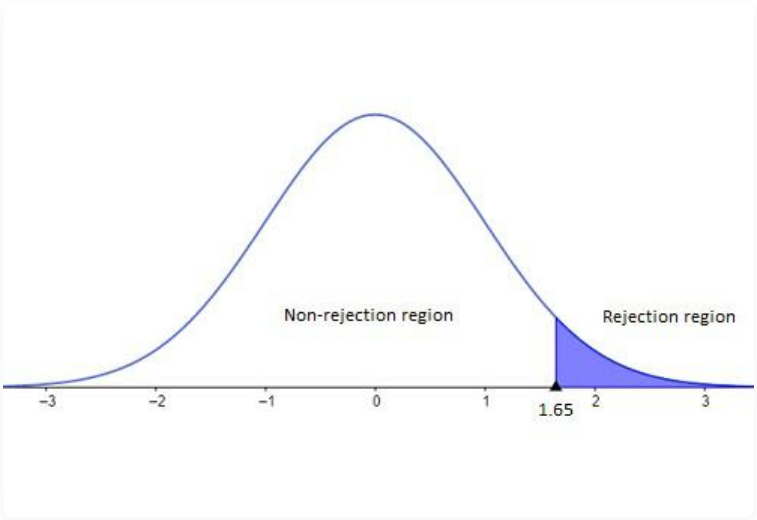
z	$\cdots$	0.05	$\cdots$
$\vdots$	$\ddots$	$\vdots$	
1.6	$\cdots$	0.95	
$\vdots$			

The area of the non-rejection region will be found in the row 1.6 and column 0.05. Taking the sum of 1.6 and 0.05, thus

$$1.6 + 0.05 = 1.65$$

Step 4

To visualize where the critical value is, the graph is represented below.



The rejection region is on the right tail of the curve because the alternative hypothesis has an inequality sign of greater than. Which means that if the test statistics is greater than the critical value, the null hypothesis will be rejected.

Step 5

Since the rejection region is in the right side of the curve, then fail to reject the null hypothesis if the test statistic is less than or equal to 1.65. The decision rule will be

$$\text{When } z > 1.65, \text{ reject } H_0$$

Step 6

c.) The test statistics that will be using is a one-sample test

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}},$$

where:

- z is the test statistics,
- $\bar{x}$ is the sample mean,
- μ is the population mean,
- σ is the population standard deviation, and
- n is the sample size.

The one-sample test should be used in the given claim because it has a one-parameter comparing to the given population mean assuming that it's normally distributed.

Step 7

By substituting the sample size n of 36, the sample mean $\bar{x}$ of 21, the population mean μ of 20, and population standard deviation σ of 5 into the formula for the test statistic of the one-sample test, thus

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{21 - 20}{5/\sqrt{36}} = 1.2.$$

In conclusion, the value of the test statistic is

$$z = 1.2$$

Step 8

d.) Since the test statistic is 1.2 which is less than the critical value 1.65, then the null hypothesis fails to be rejected. The decision will be

- At 0.05 significance level **fail** to reject H_0.

Step 9

e.) The p-value or probability value represents the probability of how likely it is to obtain the value of the test statistics given that the null hypothesis is true. To find the p-value, assume that H_0 is true. Then the standard normal distribution will be

$$P(z > 1.2).$$

Step 10

Apply complement rule in the standard normal distribution and find the area in the z-score table to find the p-value, thus

$$p = P(z > 1.2) = 1 - P(z \leq 1.2)$$

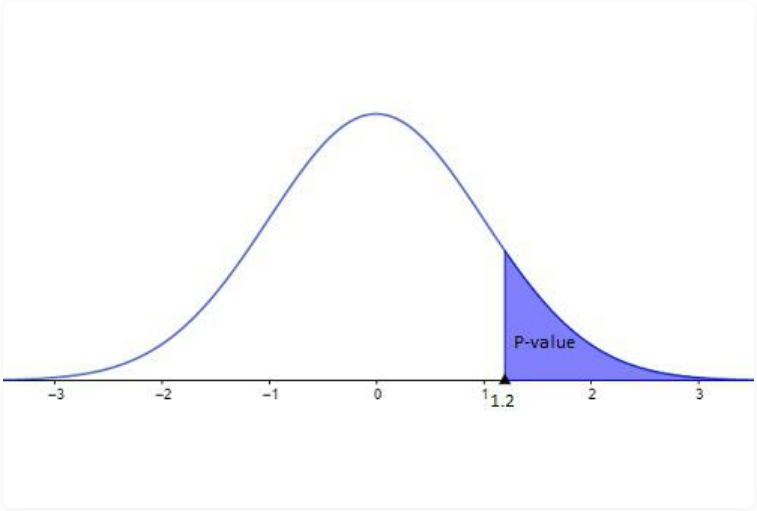
Using the z-score table to find the area of the probability $P(z \leq 1.2)$, by looking up at the table:

z	$\cdots$	0.00	$\cdots$
$\vdots$	$\ddots$	$\vdots$	
1.2	$\cdots$	0.88493	
$\vdots$			

Substitute the area of $P(z \leq 1.2)$ found in the z-table into the equation, thus

$$p = 1 - 0.88493 = 0.11507 \approx 0.1151.$$

To visualize the p-value, the graph is represented below where the p-value is the shaded region.



Step 11

In conclusion the, the p-value is

$$0.1151$$

which is having an 11.51% chance to find a z value equal to or greater than to it.

🎯 **Result**

a.) one-tailed test

b.) When z $>$ 1.65, reject H_0

c.) $z = 1.2$

d.) fail to reject H_0

e.) $p = 0.1151$

PROBLEM STATEMENT

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Step 1

Given:

$$H_0 : \mu \geq 220$$
$$H_1 : \mu < 220$$
$$\alpha = \text{Significance level} = 0.03$$
$$n = \text{Sample size} = 64$$
$$\overline{x} = \text{Sample mean} = 215$$
$$\sigma = \text{Population standard deviation} = 15$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

Step 2

(b) The rejection region of a left-tailed test with $\alpha = 0.03$ contains all z-scores below the z-score z_0 that has a probability of 0.03 to its left.

$$P(z < z_0) = 0.03$$

Let us determine the z-score that corresponds with a probability of $1 - 0.03 - 0.50 = 0.47$ in the normal probability table of the appendix (z_0 is then the negative of this z-score). We note that the probability closest to 0.47 is 0.4699, which is given in the row starting with "1.8" and in the column starting with ".08", thus the corresponding z-score is $1.8 + .08 = 1.88$.

$$z_0 = -1.88$$

Thus the rejection region then contains all z-scores smaller than -1.88.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{215 - 220}{15/\sqrt{64}} \approx -2.67$$

Step 4

(d) Result part (b): $z < -1.88$ results in the rejection of the null hypothesis H_0

Result part (c): $z = -2.67$

Since the z-score of -2.67 is less than -1.88, we reject the null hypothesis H_0

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is left-tailed, the P-value is the probability that the z-score is smaller than $z = -2.67$.

$$P = P(z < -2.67)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(z < 2.67)$ is given in the row starting with "2.6" and in the column starting with ".07" of the table.

$$\begin{aligned} P &= P(z < -2.67) \\ &= P(z > 2.67) \\ &= 1 - P(z < 2.67) \\ &= 1 - P(z < 0) - P(0 < z < 2.67) \\ &= 1 - 0.5000 - 0.4962 \\ &= 0.0038 \\ &= 0.38\% \end{aligned}$$

🎯 **Result**

(a) Left-tailed

(b) $z < -1.88$

(c) $z = -2.67$

(d) Reject H_0

(e) $0.0038 = 0.38\%$

PROBLEM STATEMENT

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Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 48$$
$$\overline{x} = \text{Sample mean} = 59500$$
$$\sigma = \text{Population standard deviation} = 5000$$

(a) Given claim: Mean differs from 60,000 miles

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. If the null hypothesis contains an equality, then the alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu = 60,000$$
$$H_a : \mu \neq 60,000$$

Step 2

(b) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right and all z-scores below the z-score $-z_0$ that has a probability of 0.05 to its left.

$$P(z < -z_0 \text{ or } z > z_0) = 0.05$$

Since the standard normal distribution is symmetric about 0:

$$P(z < -z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix. We note that the probability closest to 0.475 is 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = 1.96$$

Thus the rejection region then contains all z-scores smaller than -1.96 and all z-scores greater than 1.96.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{59500 - 60000}{5000/\sqrt{48}} \approx -0.69$$

Step 4

(d) Result part (b): $z < -1.96$ and $z > 1.96$ results in the rejection of the null hypothesis H_0

Result part (c): $z = -0.69$

Since the z-score of -0.69 is between -1.96 and 1.96, we fail to reject the null hypothesis H_0.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -0.69$ or larger than $z = 0.69$.

$$P = P(z < -0.69 \text{ or } z > 0.69)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(z < 0.69)$ is given in the row starting with "0.6" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z < -0.69 \text{ or } z > 0.69) \\ &= 2P(z > 0.69) \\ &= 2(1 - P(z < 0.69)) \\ &= 2(1 - P(z < 0) - P(0 < z < 0.69)) \\ &= 2(1 - 0.5000 - 0.2549) \\ &= 2(0.2451) \\ &= 0.4902 \\ &= 49.02\% \end{aligned}$$

Result

(a) $H_0 : \mu = 60,000$, $H_a : \mu \neq 60,000$

(b) $z < -1.96$ or $z > 1.96$

(c) $z = -0.69$

(d) Fail to reject H_0

(e) $0.4902 = 49.02\%$

PROBLEM STATEMENT

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Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 50$$
$$\overline{x} = \text{Sample mean} = 2.75$$
$$\sigma = \text{Population standard deviation} = 1$$

(a) Given claim: Mean less than 3

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \geq 3$$
$$H_a : \mu < 3$$

Step 2

(b) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score $-z_0$ that has a probability of 0.05 to its left.

$$P(z < -z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability closest to 0.45 is 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores smaller than $-z_0 = -1.65$.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{2.75 - 3}{1/\sqrt{50}} \approx -1.77$$

Step 4

(d) Result part (b): $z < -1.65$ results in the rejection of the null hypothesis H_0

Result part (c): $z = -1.77$

Since the z-score of -1.77 is between -1.65, we fail to reject the null hypothesis H_0.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.77$.

$$P = P(z < -1.77)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.77)$ is given in the row starting with "1.7" and in the column starting with ".07" of the table.

$$\begin{aligned} P &= P(z < -1.77) \\ &= P(z > 1.77) \\ &= 1 - P(z < 1.77) \\ &= 1 - P(z < 0) - P(0 < z < 1.77) \\ &= 1 - 0.5000 - 0.4616 \\ &= 0.0384 \\ &= 3.84\% \end{aligned}$$

🎯 **Result**

(a) $H_0 : \mu = 3, H_a : \mu \neq 60,000$

(b) $z < -1.65$

(c) $z = -1.77$

(d) Reject H_0

(e) $0.0384 = 3.84\%$

PROBLEM STATEMENT

Chapter 10, Page 347

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 36$$
$$\overline{x} = \text{Sample mean} = 6.2$$
$$\sigma = \text{Population standard deviation} = 0.5$$

(a) Given claim: Mean less than 6.8

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \geq 6.8$$
$$H_a : \mu < 6.8$$

Step 2

(b) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score $-z_0$ that has a probability of 0.05 to its left.

$$P(z < -z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability closest to 0.45 is 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores smaller than $-z_0 = -1.65$.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{6.2 - 6.8}{0.5/\sqrt{36}} \approx -7.20$$

Step 4

(d) Result part (b): $z < -1.65$ results in the rejection of the null hypothesis H_0

Result part (c): $z = -7.20$

Since the z-score of -7.20 is smaller than -1.65, we reject the null hypothesis H_0.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is left-tailed, the P-value is the probability that the z-score is smaller than $z = -7.20$.

$$P = P(z < -7.20)$$

Determine corresponding probability using the normal probability table in the appendix. Since $z = 7.20$ is smaller than all z-scores in the table, the probability $P(0 < z < 7.20)$ is approximately 0.5000.

$$\begin{aligned} P &= P(z < -7.20) \\ &= P(z > 0.20) \\ &= 1 - P(z < 7.20) \\ &= 1 - P(z < 0) - P(0 < z < 7.20) \\ &\approx 1 - 0.5000 - 0.5000 \\ &= 0 \\ &= 0\% \end{aligned}$$

🎯 **Result**

(a) $H_0 : \mu \geq 6.8, H_a : \mu < 6.8$

(b) $z < -1.65$

(c) $z = -7.20$

(d) Reject H_0

(e) $0 = 0\%$

PROBLEM STATEMENT

Chapter 10, Page 348

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 35$$
$$\overline{x} = \text{Sample mean} = 84.85$$
$$\sigma = \text{Population standard deviation} = 3.24$$

(a) Given claim: Mean is more than 80

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \leq 80$$
$$H_a : \mu > 80$$

Step 2

(b) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all z-scores above the z-score z_0 that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability closest to 0.49 is 0.4901 , which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than 2.33.

Step 3

(c) The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{84.85 - 80}{3.24/\sqrt{35}} \approx 8.86$$

Step 4

(d) Result part (b): $z > 2.33$ results in the rejection of the null hypothesis H_0

Result part (c): $z = 8.86$

Since the z-score of 8.86 is greater than 2.33, we reject the null hypothesis H_0.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is right-tailed, the P-value is the probability that the z-score is larger than $z = 0.69$.

$$P = P(z > 8.86)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 8.86)$ is given in the row starting with "8.8" and in the column starting with ".06" of the table. Since the table does not contain a row with "8.8," we approximate the probability as 0.5.

$$\begin{aligned} P &= P(z > 8.86) \\ &= 1 - P(z < 8.86) \\ &= 1 - P(z < 0) - P(0 < z < 8.86) \\ &\approx 1 - 0.5000 - 0.5000 \\ &= 0 \\ &= 0\% \end{aligned}$$

🎯 **Result**

(a) $H_0 : \mu \leq 80, H_a : \mu > 80$

(b) $z > 2.33$

(c) $z = 8.86$

(d) Reject H_0

(e) $0 = 0\%$

PROBLEM STATEMENT

Chapter 10, Page 352

Step 1

Given:

$$\begin{aligned} H_0 &: \mu \leq 10 \\ H_1 &: \mu > 10 \\ \alpha &= \text{Significance level} = 0.05 \\ n &= \text{Sample size} = 10 \\ \overline{x} &= \text{Sample mean} = 12 \\ s &= \text{Sample standard deviation} = 3 \end{aligned}$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 10 - 1 = 9$ and in the column with $\alpha = 0.05$.

$$t = 1.833$$

The rejection region then contains all values greater than 1.833 and thus we reject H_0 when the t-value is larger than 1.833.

Step 2

(b) Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{12 - 10}{3/\sqrt{10}} \approx 2.108$$

Step 3

(c) Result part (a): $t > 1.833$ results in the rejection of the null hypothesis H_0

Result part (b): $t = 2.108$

Since the t-value of 2.108 is larger than 1.833, we reject the null hypothesis H_0.

🎯 Result

(a) Reject H_0 when $t > 1.833$

(b) $t = 2.108$

(c) Reject H_0

PROBLEM STATEMENT

Chapter 10, Page 352

Step 1

Given:

$$
\begin{aligned}
H_0 &: \mu = 400 \\
H_1 &: \mu \neq 400 \\
\alpha &= \text{Significance level} = 0.01 \\
n &= \text{Sample size} = 12 \\
\overline{x} &= \text{Sample mean} = 407 \\
s &= \text{Sample standard deviation} = 6
\end{aligned}
$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.201$$

The rejection region then contains all values smaller than -2.201 and all values greater than 2.201,thus we reject H_0 when $t < -2.201$ or $t > 2.201$.

Step 2

(b) Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{407 - 400}{6/\sqrt{12}} \approx 4.041$$

Step 3

(c) Result part (a): $t < -2.201$ or $t > 2.201$ results in the rejection of the null hypothesis H_0

Result part (b): $t = 4.041$

Since the t-value of 4.401 is larger than 2.021, we reject the null hypothesis H_0.

🎯 Result

(a) Reject H_0 when $t < -2.201$ or $t > 2.201$.

(b) $t = 4.041$

(c) Reject H_0

PROBLEM STATEMENT

Chapter 10, Page 353

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 28$$
$$\overline{x} = \text{Sample mean} = 42$$
$$s = \text{Sample standard deviation} = 2.1$$

Given claim: Mean exceeds 40

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 40$$
$$H_a : \mu > 40$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 28 - 1 = 27$ and in the column with $\alpha = 0.05$.

$$t = 1.703$$

The rejection region then contains all values greater than 1.703 and thus we reject H_0 when the t-value is larger than 1.703.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{42 - 40}{2.1/\sqrt{28}} \approx 5.040$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t > 1.703 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean number of calls per salesperson per week is more than 40.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 353

Step 1

Given:

$$\alpha = \text{Significance level} = 0.10$$
$$n = \text{Sample size} = 24$$
$$\overline{x} = \text{Sample mean} = 40.6$$
$$s = \text{Sample standard deviation} = 2.7$$

Given claim: Mean is less than 42.3 minutes (faster)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 42.3$$
$$H_a : \mu < 42.3$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all t-value below the t-value $-t_0$ that has a probability of 0.10 to its left.

$$P(t < -t_0) = 0.10$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 24 - 1 = 23$ and in the column with $\alpha = 0.10$ (which is the negation of the critical value in the table).

$$t = -1.319$$

The rejection region then contains all values smaller than -1.319 and thus we reject H_0 when the t-value is smaller than -1.319.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{40.6 - 42.3}{2.7/\sqrt{24}} \approx -3.085$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -3.085 < -1.319 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the new method is faster.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 353

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 10$$
$$\overline{x} = \text{Sample mean} = 50000$$
$$s = \text{Sample standard deviation} = 10000$$

Given claim: Mean exceeds 40000

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 40000$$
$$H_a : \mu > 40000$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 10 - 1 = 9$ and in the column with $\alpha = 0.05$.

$$t = 1.833$$

The rejection region then contains all values greater than 1.833 and thus we reject H_0 when the t-value is larger than 1.833.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{50000 - 40000}{10000/\sqrt{10}} \approx 3.162$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 3.162 > 1.833 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the residents of Wilmington, Delaware, have more income than the national average.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 353

Solution A

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 20$$

(a) It is only appropriate to conduct a test of hypothesis, when the data values originate from a normal population.

It is safe to assume that the data values originate from a normal population when the distribution of the data values does not show strong curvature nor strong outliers.

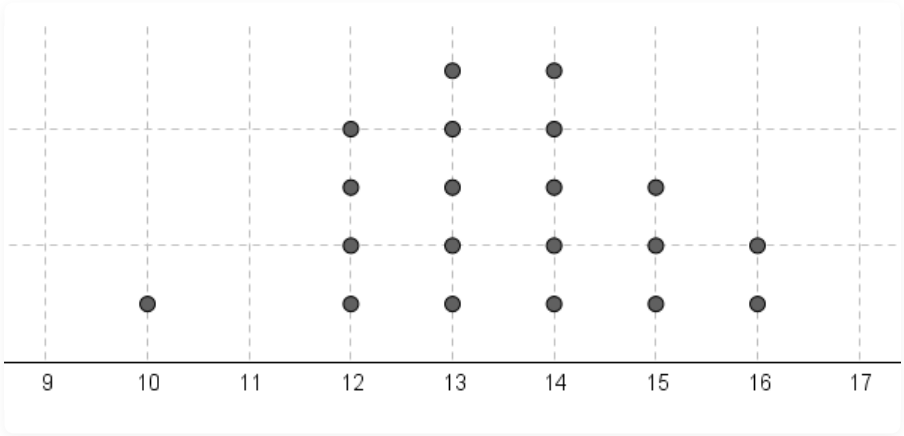
Step 2

(b)

Dotplot

Create a number line

For every given data value place a dot above the corresponding number on the number line.



We note that most of the dots lie in the middle of the dotplot, thus the distribution is approximately symmetric and bell-shaped.

Moreover, we don't note any strong outliers and thus it is safe to assume that the data values originate from a normal distribution.

Step 3

(c) Given claim: Mean is less than 15

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 15$$
$$H_a : \mu < 15$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{14 + 14 + 16 + ... + 10 + 13}{20} = \frac{270}{20} = 13.5$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(14 - 13.5)^2 + ... + (13 - 13.5)^2}{20 - 1}} \approx 1.5044$$

Step 4

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value t_0 that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 20 - 1 = 19$ and in the column with $\alpha = 0.05$ one-tailed (which is the negation of the critical value in the table).

$$t = -1.729$$

The rejection region then contains all values smaller than -1.729 and thus we reject H_0 when the t-value is smaller than -1.729.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{13.5 - 15}{1.5044/\sqrt{20}} \approx -4.459$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -4.459 < -1.729 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean number of complaints per airport is less than 15 per month.

Result

(a) Data values originate from a normal population

(b) Yes

(c) There is sufficient evidence to support the claim that the mean number of complaints per airport is less than 15 per month.

Solution B

Step 1

The problem wants us to test whether the mean number of complaints per airport is less than 15 per month. In making assumptions, let us recall the condition before conducting a hypothesis. These conditions should be checked first to make sure that the results are reliable. We will also need to make a frequency distribution table to show that the information given can be used in testing a hypothesis. Finally, if the previous questions are answered positively, we can proceed to test the hypothesis.

Step 2

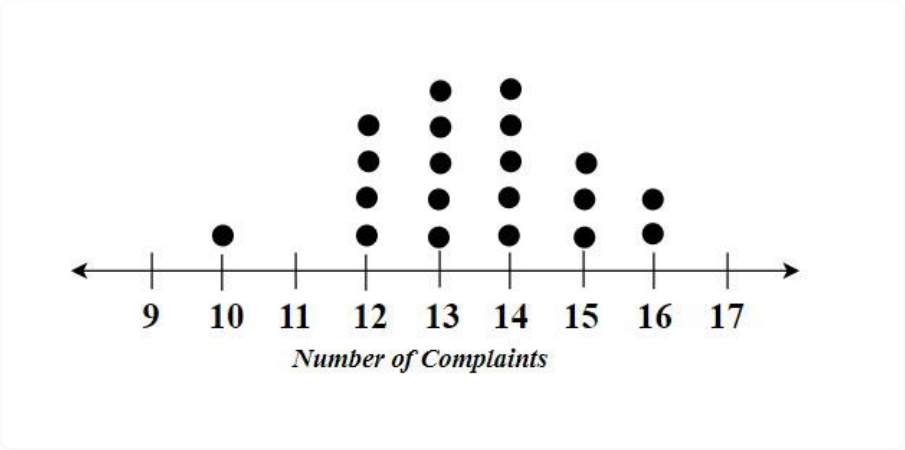
(a) Assumptions

Before conducting a test hypothesis, we need to assume that we will be working on data from a population following a normal distribution.

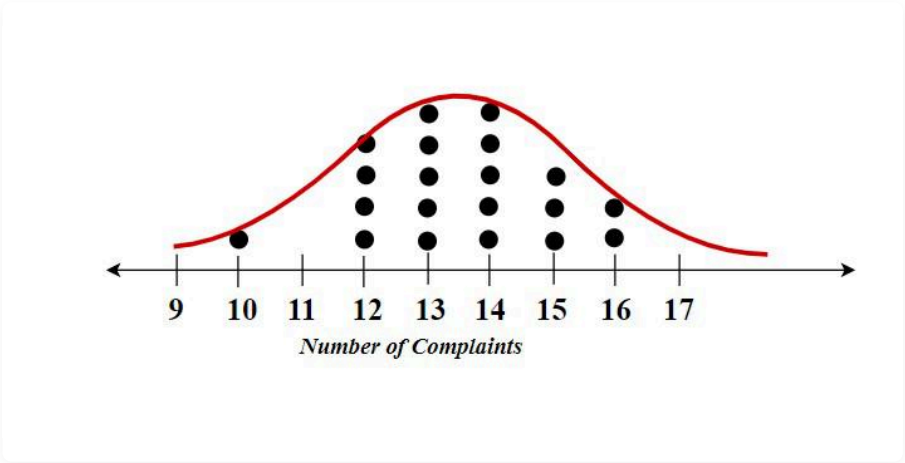
Step 3

(b) Plot of the Number of Complaints per Airport:

Here, we construct a dot plot to represent the frequency of the number of complaints per airport. In the information given, there is 1 airport with 10 complaints, 4 airports with 12 complaints each, 5 airports with 13 complaints each, 5 airports with 14 complaints each, 3 airports with 15 complaints each, and 2 airports with 16 complaints each. Showing this information in a dot plot we have:



By just looking at the obtained dot plot, the frequency distribution of the samples follows a bell curve-like pattern.



Since the random sample collected is from the population, **we can say that it is reasonable to say that the population also follows a bell curve shape which also indicates a normal distribution**.

Step 4

(c) Testing of Hypothesis:

Null and Alternate Hypothesis

Our goal is to answer whether the mean number of complaints per airport is less than 15 per month. Thus, our **null hypothesis** is that the population mean is at least 15 complaints while the **alternate hypothesis** is that the population mean is less than 15 complaints. We can express these hypotheses as:

$$H_0 : \mu \geq 15$$
$$H_1 : \mu < 15$$

Here, we will use a *one-tailed test* because we only want to know whether there has been a decrease in the number of complaints. Now, look at the inequality of the alternate hypothesis. Since it is less than, then the region of rejection is in the *left tail* of the distribution.

Step 5

Test Statistic

In part (b) we confirm that the distribution follows a normal distribution. Also, we do not know the variance of the population, thus we will use the left-tailed $t-$distribution. The value of test-statistic, when population standard deviation is unknown, can be calculated by:

$$t = \frac{\overline{x} - \mu}{\frac{s}{\sqrt{n}}}$$

where $\overline{x}$ is the sample mean, μ is the hypothesized population mean, s is the sample standard deviation, and n is the number of observations in the sample.

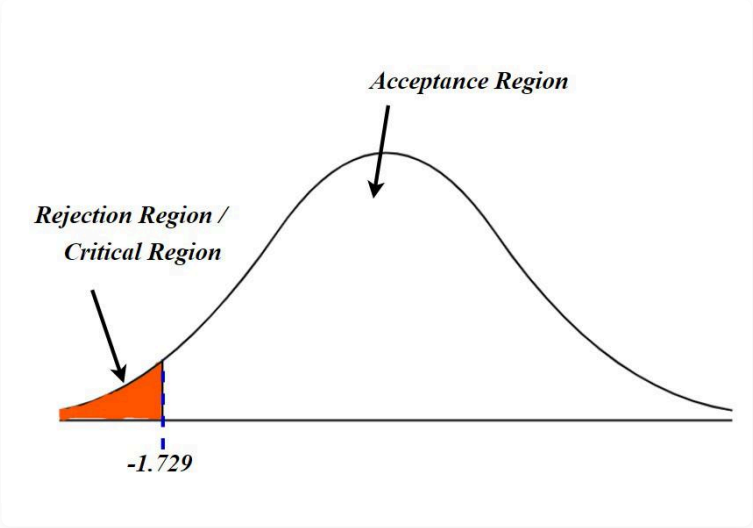
Step 6

Decision Rule

To formulate the decision rule, we need to use the table for the critical values of t for the one-tailed test. To locate the critical value t, we first determine the degrees of freedom (df). This value is the difference between the total number of observations and 1. In the information given, there are 20 samples. Thus,

$$df = 20 - 1 = 19$$

The first column of the table shows the degrees of freedom and the first row shows the confidence interval. The value at row 19 (degrees of freedom) and column 0.05 (level of confidence) is 1.729 but because our rejection region is on the left side of the distribution, then we will take the negative of the value shown. Hence, our critical region is -1.729. Therefore, **the decision rule is to reject H_0 if the value of t is less than -1.729**.



Step 7

Decision

Now that we have a decision rule, we proceed to calculating t. We know that our $n = 20$ for there are 20 observations, and the hypothesized population mean is 15 then $\mu = 15$. We do not know the sample mean and the sample standard deviation yet so we compute for the two values using the formulas:

$$\overline{x} = \frac{\Sigma x}{n} \qquad\qquad s = \sqrt{\frac{\Sigma (x - \overline{x})^2}{n-1}}$$

Sample Mean:

$$\overline{x} = \frac{\Sigma x}{n} = \frac{14 + 14 + 16 + \cdots + 13 + 10 + 13}{20} = \frac{270}{20} = 13.5$$

Sample Standard Deviation:

$$\begin{aligned}
s &= \sqrt{\frac{\Sigma (x - \overline{x})^2}{n-1}} \\
&= \sqrt{\frac{(14 - 13.5)^2 + (14 - 13.5)^2 + \cdots + (10 - 13.5)^2 + (13 - 13.5)^2}{20 - 1}} \\
&= \sqrt{\frac{(0.5)^2 + (0.5)^2 + \cdots + (-3.5)^2 + (-0.5)^2}{19}} \\
&= \sqrt{\frac{0.25 + 0.25 + 12.25 + 0.25}{19}} \\
&= \sqrt{\frac{43}{19}} \\
&= \sqrt{2.2632} \\
&= 1.5044
\end{aligned}$$

Step 8

We can now proceed to calculating t by substituting $\overline{x} = 13.5, \mu = 15, s = 1.5044, n = 20$ into the t formula:

$$
\begin{aligned}
t &= \frac{\overline{x} - \mu}{\frac{s}{\sqrt{n}}} && \text{(Write the formula.)} \\
&= \frac{13.5 - 15}{\frac{1.5044}{\sqrt{20}}} && \text{(Substitute the values.)} \\
&= \frac{-1.5}{\frac{1.5044}{4.4721}} && \text{(Evaluate the square root.)} \\
&\approx -4.4590 && \text{(Simplify.)}
\end{aligned}
$$

Since the computed t value is -4.4590, then we reject the null hypothesis because -4.4590 is less than the critical value -1.729 and -4.4590 falls into the rejection region.

$$\text{Decision: } t = -4.4590 < -1.729 \longrightarrow \text{Reject } H_0$$

We've just rejected the null hypothesis which says that the population mean is at least 15. Thus, we will accept the alternate hypothesis which says that the population mean is less than 15. This means that the average number of complaints per airport is less than 15 per month. The test result could be a piece of good news to the administrators since it does not exceed 15 complaints but it is still a complaint regardless of the number. Thus, administrators still need to improve the airports' electronic ticketing services.

PROBLEM STATEMENT

Chapter 10, Page 355

Step 1

Given:

$$H_0 : \mu \geq 20$$
$$H_1 : \mu < 20$$
$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 5$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 5 - 1 = 4$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -3.747$$

The rejection region then contains all values smaller than -3.747 and thus we reject H_0 when the t-value is smaller than -3.747.

Step 2

(b) The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{18 + 15 + 12 + 19 + 21}{5} = \frac{85}{5} = 17$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(18-17)^2 + (15-17)^2 + (12-17)^2 \\ +(19-17)^2 + (21-17)^2}{5-1}} \approx 3.5355$$

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{17 - 20}{3.5355/\sqrt{5}} \approx -1.897$$

Step 3

(c) By part (a): Reject H_0 when $t < -3.747$

By part (b): $t = -1.897$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -1.897 > -3.747 \Rightarrow \text{ Fail to reject } H_0$$

Step 4

(d) The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 1.897 in the row $df = n - 1 = 5 - 1 = 4$ (one-tailed):

$$0.05 < P < 0.10$$

Note: When using technology, then you obtain the P-value 0.06535.

🎯 **Result**

(a) Reject H_0 when $t < -3.747$

(b) $t = -1.897$

(c) Fail to reject H_0

(d) $0.05 < P < 0.10$

PROBLEM STATEMENT

Chapter 10, Page 355

Step 1

Given:

$$H_0 : \mu = 100$$
$$H_1 : \mu \neq 100$$
$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 6$$

(a) If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 6 - 1 = 5$ and in the column with $\alpha = 0.05$.

$$t_0 = 2.571$$

The rejection region then contains all values smaller than -2.571 and all values greater than 2.571, thus we reject H_0 when $t < -2.571$ or $t > 2.571$.

Step 2

(b) The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}118+105+112\\+119+105+111\end{array}}{6} = \frac{670}{6} \approx 111.6667$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{\begin{array}{l}(118-111.6667)^2+(105-111.6667)^2\\+(112-111.6667)^2+(119-111.6667)^2\\+(105-111.6667)^2+(111-111.6667)^2\end{array}}{6-1}} \approx 6.0553$$

Determine the value of the test statistic:

$$t = \frac{\overline{x}-\mu_0}{s/\sqrt{n}} = \frac{111.6667-100}{6.0553/\sqrt{6}} \approx 4.719$$

Step 3

(c) By part (a): Reject H_0 when $t < -2.571$ or $t > 2.571$

By part (b): $t = 4.719$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 4.719 > 2.571 \Rightarrow \text{ Reject } H_0$$

Step 4

(d) The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 4.719 in the row $df = n-1 = 6-1 = 5$ (two-tailed):

$$0.001 < P < 0.01$$

Note: When using technology, then you obtain the P-value 0.00524.

🎯 **Result**

(a) Reject H_0 when $t < -2.571$ or $t > 2.571$

(b) $t = 4.719$

(c) Reject H_0

(d) $0.001 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 355

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 10$$

Given claim: Mean exceeds 1.4 (increased)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 1.4$$
$$H_a : \mu > 1.4$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}1.5 + 1.6 + 1.5 + 1.4 + 1.9 \\ +1.4 + 1.3 + 1.9 + 1.8 + 1.7\end{array}}{10} = \frac{16}{10} = 1.6$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(1.5 - 1.6)^2 + (1.6 - 1.6)^2 + ... + (1.7 - 1.6)^2}{10 - 1}} \approx 0.2160$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 10 - 1 = 9$ and in the column with $\alpha = 0.01$.

$$t = 3.250$$

The rejection region then contains all values greater than 3.250 and thus we reject H_0 when the t-value is larger than 3.250.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{1.6 - 1.4}{0.2160/\sqrt{10}} \approx 2.928$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 2.928 < 3.250 \Rightarrow \text{ Fail to reject } H_0$$

There is no sufficient evidence to support the claim that water consumption has increased.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 2.928 in the row $df = n - 1 = 10 - 1 = 9$ (one-tailed):

$$0.001 < P < 0.01$$

Note: When using technology, then you obtain the P-value 0.00841.

🎯 **Result**

Yes, $0.001 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 355

Step 1

Given:

$$\alpha = \text{Significance level} = 0.025$$
$$n = \text{Sample size} = 9$$

Given claim: Mean exceeds 2160 (increased)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 2160$$
$$H_a : \mu > 2160$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}2159 + 2170 + 2180 \\ +2179 + 2160 + 2167 \\ +2171 + 2181 + 2185\end{array}}{9} = \frac{19552}{9} \approx 2172.4444$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(2159 - 2172.4444)^2 + ... + (2185 - 2172.4444)^2}{9 - 1}} \approx 9.3822$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.025$ contains all t-value above the t-value t_0 that has a probability of 0.025 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 9 - 1 = 8$ and in the column with $\alpha = 0.05$.

$$t = 2.752$$

The rejection region then contains all values greater than 2.752 and thus we reject H_0 when the t-value is larger than 2.752.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{2172.4444 - 2160}{9.3822/\sqrt{9}} \approx 3.979$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 3.979 > 2.752 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that Holdlonger has increased the shelf life of chlorine.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 3.979 in the row $df = n - 1 = 9 - 1 = 8$ (one-tailed):

$$0.001 < P < 0.01$$

Note: When using technology, then you obtain the P-value 0.00203.

🎯 **Result**

Yes, $0.001 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 356

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 12$$

Given claim: Mean exceeds 50

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 50$$
$$H_a : \mu > 50$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{51 + 175 + 47 + ... + 42 + 100}{12} = \frac{990}{12} = 82.5$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(51 - 82.5)^2 + (175 - 82.5)^2 + ... + (100 - 82.5)^2}{12 - 1}} \approx 59.4926$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.05$.

$$t = 2.201$$

The rejection region then contains all values greater than 2.201 and thus we reject H_0 when the t-value is larger than 2.201.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{82.5 - 50}{59.4926/\sqrt{12}} \approx 1.892$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 1.892 < 2.201 \Rightarrow \text{ Fail to reject } H_0$$

There is no sufficient evidence to support the claim that the mean number is greater than 50.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 1.892 in the row $df = n - 1 = 12 - 1 = 11$ (one-tailed):

$$0.025 < P < 0.05$$

Note: When using technology, then you obtain the P-value 0.04255.

🎯 **Result**

No, $0.025 < P < 0.05$

PROBLEM STATEMENT

Chapter 10, Page 356

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 15$$

Given claim: Mean exceeds 53

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 53$$
$$H_a : \mu > 53$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{53 + 57 + 50 + ... + 59 + 56}{15} = \frac{846}{15} = 56.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(53 - 56.4)^2 + (57 - 56.4)^2 + ... + (56 - 56.4)^2}{20 - 1}} \approx 3.7378$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 15 - 1 = 14$ and in the column with $\alpha = 0.05$.

$$t = 2.145$$

The rejection region then contains all values greater than 2.145 and thus we reject H_0 when the t-value is larger than 2.145.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{56.4 - 53}{3.7378/\sqrt{15}} \approx 3.523$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 3.523 > 2.145 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean number of interviews conducted by the agents is more than 53 per week.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 3.523 in the row $df = n - 1 = 15 - 1 = 14$ (one-tailed):

$$0.001 < P < 0.01$$

Note: When using technology, then you obtain the P-value 0.00169.

🎯 **Result**

Yes, $0.001 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 359

Step 1

Given:

$$H_0 : \pi \leq 0.70$$
$$H_1 : \pi > 0.70$$
$$n = \text{Sample size} = 100$$
$$\hat{p} = \text{Sample proportion} = 0.75$$
$$\alpha = \text{Significance level} = 0.05$$

(a) If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed test}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Step 2

(b) Determine the value of the test-statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} = \frac{0.75 - 0.70}{\sqrt{\frac{0.70(1 - 0.70)}{100}}} \approx 1.09$$

Step 3

(c) Result part (a): Reject H_0 when $z > 1.65$

Result part (b): $z = 1.09$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 1.09 < 1.65 \Rightarrow \text{Fail to reject } H_0$$

🎯 Result

(a) Reject H_0 when $z > 1.65$

(b) $z = 1.09$

(c) Fail to reject H_0

PROBLEM STATEMENT

Chapter 10, Page 359

Step 1

Given:

$$H_0 : \pi = 0.40$$
$$H_1 : \pi \neq 0.40$$
$$n = \text{Sample size} = 120$$
$$\hat{p} = \text{Sample proportion} = 0.30$$
$$\alpha = \text{Significance level} = 0.05$$

(a) If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed test}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right and all z-scores below the z-score $-z_0$ that has a probability of 0.05 to its left.

$$P(z < -z_0 \text{ or } z > z_0) = 0.05$$

Since the standard normal distribution is symmetric about 0:

$$P(z < -z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix. We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = 1.96$$

Thus the rejection region then contains all z-scores smaller than -1.96 and all z-scores greater than 1.96 and thus we reject H_0 when the z-score is less than -1.96 or greater than 1.96.

Step 2

(b) Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.30 - 0.40}{\sqrt{\frac{0.40(1-0.40)}{210}}} \approx -2.96$$

Step 3

(c) Result part (a): Reject H_0 when $z - 1.96$ or $z > 1.96$

Result part (b): $z = -2.96$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -2.96 < -1.96 \Rightarrow \text{ Reject } H_0$$

🎯 Result

(a) Reject H_0 when $z - 1.96$ or $z > 1.96$

(b) $z = -2.96$

(c) Reject H_0

PROBLEM STATEMENT

Chapter 10, Page 359

Solution A

Step 1

First step is to state the null hypothesis and the alternative hypothesis. It asked to conclude that a larger proportion of men were driving on the New Jersey Turnpike π than the national statistics indicate which is 0.52, thus, it will be the alternate hypothesis H_1. Then the null hypothesis H_0 be the opposite of the statement, it will be the smaller or equal proportion of men were driving on the New Jersey Turnpike than the national statistics indicate. The hypothesis will be

$$H_0 : \pi \leq 0.52$$
$$H_1 : \pi > 0.52$$

Step 2

Second step is to set the significance level. Based on the given, the significance level will be at

$$\alpha = 0.01$$

Step 3

Third step is to select a test statistic, since the problem is in the assumption of non-parametric test and it needs to compare the proportion of men driving in Turnpike and the national statistics indicate. Thus, it needed to use the one proportion as testing for the hypothesis. The test statistic formula will be

$$z = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

where:

- p is the sample proportion,
- π is the population proportion, and
- z is the test statistic

Step 4

Fourth step is to formulate the decision rule. In order to accomplish, it needed to find the critical value. Since the curve is one-tailed where the rejection region is on the right side of the curve, then the total area for the right side is 0.5000 subtracting it by the significance level α is 0.01. Thus,

$$0.5000 - 0.01 = 0.49$$

Next is to look 0.49 or closes to it on the table of the area under the normal curve. It will be:

z	$\cdots$	0.03	$\cdots$
$\vdots$	$\ddots$	$\vdots$	
2.3	$\cdots$	0.4901	
$\vdots$			

The row will be 2.3 and the column is 0.03, adding the row and the column to find the critical value. Thus,

$$\begin{aligned} z &= 2.3 + 0.03 \\ &= 2.33 \end{aligned}$$

Since the critical value is obtained, the null hypothesis is fail to be rejected when the test statistics is less than or equal to the critical value 2.33. The decision rule will be

$$\text{When z} > 2.33, \text{ reject } H_0.$$

Step 5

Fifth step is to make a decision. To accomplish it, compute the test statistic by substituting the population proportion π is 52%, sample size n of 300, sample proportion of 0.5667 into the formula of one-sample proportion test. Thus,

$$\begin{aligned} z &= \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} \\ &= \frac{0.5667 - 0.52}{\sqrt{\frac{0.52(1-0.52)}{300}}} \\ &= 1.61903119 \approx 1.62. \end{aligned}$$

Step 6

Since the test statistic is less than the critical value

$$1.62 < 2.33.$$

Then the null hypothesis is fail to be rejected. The decision will be

$$\text{Fail to reject } H_0.$$

Step 7

Sixth step is to interpret the result. Based on the result of the test, it suggested that it cannot be concluded that the proportion of men driving on the Turnpike is larger than the national statistics indicate which is 0.52.

Solution B

Step 1

Given:

$$n = \text{Sample size} = 300$$
$$x = \text{Number of successes} = 170$$
$$\hat{p} = \text{Sample proportion} = 0.62$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: π is greater than 52% (larger proportion)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 52\% = 0.52$$
$$H_1 : \pi > 0.52$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than 2.33 and thus we reject H_0 when the z-score is greater than 2.33.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{170}{300} \approx 0.5667$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.5667 - 0.52}{\sqrt{\frac{0.52(1-0.52)}{300}}} \approx 1.62$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 1.62 < 2.33 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that a larger proportion of men were driving on the New Jersey Turnpike than the national statistics indicate.

Result

No

PROBLEM STATEMENT

Chapter 10, Page 359

Step 1

Given:

$$n = \text{Sample size} = 200$$
$$x = \text{Number of successes} = 80$$
$$\alpha = \text{Significance level} = 0.02$$

Given claim: π is greater than 1 in 3 (larger population)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq \frac{1}{3} \approx 0.3333$$
$$H_1 : \pi > 0.3333$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.02$ contains all z-scores above the z-score z_0 that has a probability of 0.02 to its right.

$$P(z > z_0) = 0.02$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.02 = 0.48$ in the normal probability table of the appendix. We note that the probability 0.48 is closest to 0.4798, which is given in the row starting with "2.0" and in the column starting with ".05", thus the corresponding z-score is $2.0 + .05 = 2.05$.

$$z_0 = 2.05$$

Thus the rejection region then contains all z-scores greater than 2.05 and thus we reject H_0 when the z-score is greater than 2.05.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{80}{200} = 0.4$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.4 - 0.3333}{\sqrt{\frac{0.3333(1-0.3333)}{200}}} \approx 2.00$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 2.00 < 2.05 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that a larger proportion of students at your school have jobs.

🎯 Result

No

PROBLEM STATEMENT

Chapter 10, Page 359

Step 1

Given:

$$n = \text{Sample size} = 100$$
$$x = \text{Number of successes} = 82$$
$$\alpha = \text{Significance level} = 0.10$$

Given claim: proportion is less than 90%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \geq 90\% = 0.90$$
$$H_1 : \pi < 0.90$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all z-scores below the z-score z_0 that has a probability of 0.10 to its left.

$$P(z < z_0) = 0.10$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.10 = 0.40$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.40 is closest to 0.3997, which is given in the row starting with "1.2" and in the column starting with ".08", thus the corresponding z-score is $1.2 + .08 = 1.28$.

$$z_0 = -1.28$$

Thus the rejection region then contains all z-scores smaller than -1.28 and thus we reject H_0 when the z-score is smaller than -1.28.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{82}{100} = 0.82$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.82 - 0.9}{\sqrt{\frac{0.9(1-0.9)}{100}}} \approx -2.67$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -2.67 < -1.28 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that less than 90 percent of the orders are delivered in less than 10 minutes.

Result

Yes

Chapter 10 Exercise 26

Step 1

4. Step & Result Cards Not found

PROBLEM STATEMENT

Chapter 10, Page 359

Step 1

Given:

$$n = \text{Sample size} = 300$$
$$x = \text{Number of successes} = 170$$
$$\hat{p} = \text{Sample proportion} = 0.62$$
$$\alpha = \text{Significance level} = 5\% = 0.05$$

Given claim: π is less than 50% (decrease in proportion)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \geq 50\% = 0.50$$
$$H_1 : \pi < 0.50$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all z-scores below the z-score z_0 that has a probability of 0.10 to its left.

$$P(z < z_0) = 0.10$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than -1.65 and thus we reject H_0 when the z-score is smaller than -1.65.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{48}{100} = 0.48$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.48 - 0.50}{\sqrt{\frac{0.50(1-0.50)}{100}}} \approx -0.40$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -0.40 > -1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim that there has been a significant decrease in the proportion of students who change their major after the first year in this program

Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 362

Step 1

Given:

$$\begin{aligned}\alpha &= \text{Significance level} = 0.05\\ n &= \text{Sample size} = 100\\ \overline{X}_c &= \text{Sample mean} = 9922\\ \sigma &= \text{Population standard deviation} = 400\\ \mu_1 &= 9880\end{aligned}$$

The sampling distribution of the sample mean has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{X}_c - \mu_1}{\sigma/\sqrt{n}} = \frac{9922 - 9880}{400/\sqrt{100}} = 1.05$$

The type II error is the probability of failing to reject the null hypothesis H_0, when the alternative hypothesis H_1 is true.

More precisely, the type II error is the probability of obtaining a z-score that is more extreme than $z = 1.05$, which is thus the probability that the z-score is larger than $z = 1.05$.

$$P(\text{Type II error}) = P(z > 1.05)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.05)$ is given in the row starting with "1.0" and in the column starting with ".05" of the table.

$$\begin{aligned}P(\text{Type II error}) &= P(z > 1.05)\\ &= 1 - P(z < 1.05)\\ &= 1 - P(z < 0) - P(0 < z < 1.05)\\ &= 1 - 0.5000 - 0.3531\\ &= 0.1469\end{aligned}$$

🎯 **Result**

$$P(\text{Type II error}) = 0.1469$$

PROBLEM STATEMENT

Chapter 10, Page 362

Step 1

Given:

$$\begin{aligned}\alpha &= \text{Significance level} = 0.05\\ n &= \text{Sample size} = 100\\ \overline{X}_c &= \text{Sample mean} = 9922\\ \sigma &= \text{Population standard deviation} = 400\\ \mu_1 &= 9940\end{aligned}$$

The sampling distribution of the sample mean has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{X}_c - \mu_1}{\sigma/\sqrt{n}} = \frac{9922 - 9940}{400/\sqrt{100}} = -0.45$$

The type II error is the probability of failing to reject the null hypothesis H_0, when the alternative hypothesis H_1 is true.

More precisely, the type II error is the probability of obtaining a z-score that is more extreme than $z = -0.45$, which is thus the probability that the z-score is larger than $z = -0.45$.

$$P(\text{Type II error}) = P(z > -0.45)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 0.45)$ is given in the row starting with "0.4" and in the column starting with ".05" of the table.

$$\begin{aligned}P(\text{Type II error}) &= P(z > -0.45)\\ &= P(z < 0.45)\\ &= P(z < 0) + P(0 < z < 0.45)\\ &= 0.5000 + 0.1736\\ &= 0.6736\end{aligned}$$

🎯 **Result**

$$P(\text{Type II error}) = 0.6736$$

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.10$$
$$n = \text{Sample size} = 120$$
$$\overline{x} = \text{Sample mean} = 45500$$
$$\sigma = \text{Population standard deviation} = 3000$$

Given claim: Mean differs from 45000

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu = 45000$$
$$H_a : \mu \neq 45000$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.10$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right and all z-scores below the z-score $-z_0$ that has a probability of 0.05 to its left.

$$P(z < -z_0 \text{ or } z > z_0) = 0.10$$

Since the standard normal distribution is symmetric about 0:

$$P(z < -z_0) = P(z > z_0) = \frac{0.10}{2} = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability closest to 0.45 is 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores smaller than -1.65 and all z-scores greater than 1.65.

Step 2

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{45500 - 45000}{3000/\sqrt{120}} \approx 1.83$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 1.83 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean income is not equal to $\$45,000$.

Step 3

P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.83$ or larger than $z = 1.83$.

$$P = P(z < -1.83 \text{ or } z > 1.83)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.83)$ is given in the row starting with "1.8" and in the column starting with ".03" of the table.

$$\begin{aligned} P &= P(z < -1.83 \text{ or } z > 1.83) \\ &= 2P(z > 1.83) \\ &= 2(1 - P(z < 1.83)) \\ &= 2(1 - P(z < 0) - P(0 < z < 1.83)) \\ &= 2(1 - 0.5000 - 0.4664) \\ &= 2(0.0336) \\ &= 0.0672 \\ &= 6.72\% \end{aligned}$$

🎯 **Result**

Reasonable, $P = 0.0672 = 6.72\%$

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 10$$
$$\sigma = \text{Population standard deviation} = 3$$

Given claim: Mean is less than 50

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \geq 50$$
$$H_a : \mu < 50$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all z-scores below the z-score z_0 that has a probability of 0.01 to its left.

$$P(z < z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = -2.33$$

Thus the rejection region then contains all z-scores smaller than -2.33 and thus we reject H_0 when the z-score is smaller than -2.33.

Step 2

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}45.6 + 47.7 + 47.6 + 46.3 + 46.2 \\ +47.4 + 49.2 + 55.8 + 47.5 + 48.5\end{array}}{10} = \frac{481.8}{10} = 48.18$$

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{48.18 - 50}{3/\sqrt{10}} \approx -1.92$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -1.92 > -2.33 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean weight of the bags is less than 50 pounds.

Step 3

(b) It is appropriate to use the z distribution as the test statistic, because we have been given that the population has a normal probability distribution and the population standard deviation is known.

When the population standard deviation is unknown, then we generally need to use the t distributio

Step 4

(c) Result part (a): $z = -1.92$

P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.92$ or larger than $z = 1.92$.

$$P = P(z < -1.92 \text{ or } z > 1.92)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.92)$ is given in the row starting with "1.9" and in the column starting with ".02" of the table.

$$\begin{aligned} P &= P(z < -1.92 \text{ or } z > 1.92) \\ &= 2P(z > 1.92) \\ &= 2(1 - P(z < 1.92)) \\ &= 2(1 - P(z < 0) - P(0 < z < 1.92)) \\ &= 2(1 - 0.5000 - 0.4726) \\ &= 2(0.0274) \\ &= 0.0548 \\ &= 5.48\% \end{aligned}$$

Result

(a) No

(b) Population distribution is known to be normal and population standard deviation is known

(c) $0.0548 = 5.48\%$

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 50$$
$$\overline{x} = \text{Sample mean} = 9$$
$$\sigma = \text{Population standard deviation} = 2.8$$

Given claim: Mean is less than 10

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \geq 10$$
$$H_a : \mu < 10$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score z_0 that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than -1.65 and thus we reject H_0 when the z-score is smaller than -1.65.

Step 2

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{9 - 10}{2.8/\sqrt{50}} \approx -2.53$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -2.53 < -1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that those joining Weight Reducers on average will lose less than 10 pounds.

Step 3

P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -2.53$.

$$P = P(z < -2.53)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 2.53)$ is given in the row starting with "2.5" and in the column starting with ".03" of the table.

$$\begin{aligned} P &= P(z < -2.53) \\ &= P(z > 2.53) \\ &= 1 - P(z < 2.53) \\ &= 1 - P(z < 0) - P(0 < z < 2.53) \\ &= 1 - 0.5000 - 0.4943 \\ &= 0.0057 \\ &= 0.57\% \end{aligned}$$

🎯 Result

Yes, $P = 0.0057 = 0.57\%$

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 5\% = 0.05$$
$$n = \text{Sample size} = 50$$
$$\overline{x} = \text{Sample mean} = 16.05$$
$$\sigma = \text{Population standard deviation} = 0.03$$

Given claim: Mean is greater than 16

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \leq 16$$
$$H_a : \mu > 16$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Step 2

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{16.05 - 16}{0.03/\sqrt{50}} \approx 11.79$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 11.79 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean weight is greater than 16 ounces.

Step 3

P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is larger than $z = 11.79$.

$$P = P(z < -11.79 \text{ or } z > 11.79)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 11.79)$ is given in the row starting with "11.7" and in the column starting with ".09" of the table. However, we note that the table does not contain a row starting with "11.7", thus we estimate the probability as about 0.5000.

$$\begin{aligned} P &= P(z > 11.79) \\ &= 1 - P(z < 11.79) \\ &= 1 - P(z < 0) - P(0 < z < 11.79) \\ &\approx 1 - 0.5000 - 0.5000 \\ &= 0 \\ &= 0\% \end{aligned}$$

Result

Yes, $P = 0 = 0\%$

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\begin{aligned} \alpha &= \text{Significance level} = 0.05 \qquad (Assumption) \\ n &= \text{Sample size} = 50 \\ \overline{x} &= \text{Sample mean} = 6.8 \\ s &= \text{Sample standard deviation} = 0.9 \end{aligned}$$

Given claim: Mean is less than 7

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$\begin{aligned} H_0 &: \mu \geq 7 \\ H_a &: \mu < 7 \end{aligned}$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all t-value below the t-value $-t_0$ that has a probability of 0.10 to its left.

$$P(t < -t_0) = 0.10$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 50 - 1 = 49$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.677$$

The rejection region then contains all values smaller than -1.677 and thus we reject H_0 when the t-value is smaller than -1.677.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{6.8 - 7}{0.9/\sqrt{50}} \approx -1.571$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -1.571 > -1.677 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that students at West Virginia sleep less than the typical American.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 1.571 in the row $df = n - 1 = 50 - 1 = 49$ (one-tailed):

$$0.05 < P < 0.10$$

Note: When using technology, then you obtain the P-value 0.06131.

🎯 Result

Not reasonable, $0.05 < P < 0.10$

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.10$$
$$n = \text{Sample size} = 100$$
$$\overline{x} = \text{Sample mean} = 94$$
$$s = \text{Sample standard deviation} = 22$$

Given claim: Mean exceeds 90 (increase)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 90$$
$$H_a : \mu > 90$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.10$ contains all t-value above the t-value t_0 that has a probability of 0.10 to its right.

$$P(t > t_0) = 0.10$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 100 - 1 = 99$ and in the column with $\alpha = 0.10$.

$$t = 1.290$$

The rejection region then contains all values greater than 1.290 and thus we reject H_0 when the t-value is larger than 1.290.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{94 - 90}{22/\sqrt{100}} \approx 1.818$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 1.818 > 1.290 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to suppose the claim that there has been an increase in selling time.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 25$$
$$\overline{x} = \text{Sample mean} = 2.86$$
$$s = \text{Sample standard deviation} = 1.20$$

Given claim: Mean is less than 3.13

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 3.13$$
$$H_a : \mu < 3.13$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 25 - 1 = 24$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.711$$

The rejection region then contains all values smaller than -1.711 and thus we reject H_0 when the t-value is smaller than -1.711.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{2.86 - 3.13}{1.20/\sqrt{25}} \approx -1.125$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -1.125 > -1.711 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean number of residents in the retirement community household is less than 3.13 persons.

🎯 Result

Not reasonable

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 60$$
$$\overline{x} = \text{Sample mean} = 37.8$$
$$s = \text{Sample standard deviation} = 12.2$$

Given claim: Mean is less than 40.0 (too large)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 40$$
$$H_a : \mu < 40$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 60 - 1 = 59$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.671$$

The rejection region then contains all values smaller than -1.671 and thus we reject H_0 when the t-value is smaller than -1.671.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{37.8 - 40.00}{12.2/\sqrt{60}} \approx -1.397$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -1.397 > -1.671 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean amount of leasure time of 40.0 hours is too large and thus the information in the article appears to be true.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 364

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 10$$

Given claim: Mean is greater 14 percent

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 40$$
$$H_a : \mu > 40$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}14.6 + 16.7 + 17.4 + 17.0 + 17.8 \\ +158.4 + 13.1 + 15.8 + 14.3 + 14.5\end{array}}{10} = \frac{156.6}{10} = 15.66$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(14.6 - 15.66)^2 + ... + (14.5 - 15.66)^2}{10 - 1}} \approx 1.5436$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 10 - 1 = 9$ and in the column with $\alpha = 0.01$.

$$t = 2.821$$

The rejection region then contains all values greater than 2.821 and thus we reject H_0 when the t-value is larger than 2.821.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{15.66 - 14}{1.5436/\sqrt{10}} \approx 3.401$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 3.401 > 2.821 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean rate charged is greater than 14 percent.

Result

Reasonable

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 5$$

Given claim: Mean is less than 6

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 6$$
$$H_a : \mu < 6$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}4.8 + 5.3 + 6.5 + 4.8 \\ +6.1 + 5.8 + 6.2 + 5.6\end{array}}{8} = \frac{45.1}{8} = 5.6375$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(4.8 - 5.6375)^2 + \ldots + (5.6 - 5.6375)^2}{8 - 1}} \approx 0.6346$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 8 - 1 = 7$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -2.998$$

The rejection region then contains all values smaller than -2.998 and thus we reject H_0 when the t-value is smaller than -2.998.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{5.6375 - 6}{0.6346/\sqrt{8}} \approx -1.616$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -1.616 > -2.998 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the 30-year mortgage rate for small banks is less than 6 percent.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 1.616 in the row $df = n - 1 = 8 - 1 = 7$ (one-tailed):

$$0.05 < P < 0.10$$

Note: When using technology, then you obtain the P-value 0.07506.

🎯 **Result**

No, $0.05 < P < 0.10$

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 12$$

Given claim: Mean differs from 3.1

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu = 3.1$$
$$H_a : \mu \neq 3.1$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}3.1 + 3.3 + 3.5 + 2.6 + 2.6 + 4.3 \\ +4.4 + 3.8 + 3.1 + 4.1 + 3.1 + 3.2\end{array}}{12} = \frac{41.1}{12} = 3.425$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(3.1 - 3.425)^2 + ... + (3.2 - 3.425)^2}{12 - 1}} \approx 0.6077$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.201$$

The rejection region then contains all values smaller than -2.201 and all values greater than 2.201,thus we reject H_0 when $t < -2.201$ or $t > 2.201$.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{3.425 - 3.1}{0.6077/\sqrt{12}} \approx 1.853$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$-2.201 < 1.853 < 2.201 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a different between the national average and the sample mean from senior citizens.

Result

No

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 15$$

Given claim: Mean is more than 25

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 25$$
$$H_a : \mu > 25$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}25 + 27 + 25 + 26 + 25 \\ +28 + 28 + 27 + 24 + 26 \\ +25 + 29 + 25 + 27 + 24\end{array}}{15} = \frac{391}{15} \approx 26.0667$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(25 - 26.0667)^2 + ... + (24 - 26.0667)^2}{15 - 1}} \approx 1.5337$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 15 - 1 = 14$ and in the column with $\alpha = 0.01$.

$$t = 2.624$$

The rejection region then contains all values greater than 2.624 and thus we reject H_0 when the t-value is larger than 2.624.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{26.0667 - 25}{1.5337/\sqrt{15}} \approx 2.694$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 2.694 > 2.624 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean number of patients per day is more than 25.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 2.694 in the row $df = n - 1 = 15 - 1 = 14$ (one-tailed):

$$0.005 < P < 0.01$$

When the population mean is 25, then the chance of obtaining a sample mean of 26.0667 or higher is between 0.5% and 1%.

Note: When using technology, then you obtain the P-value 0.00873.

🎯 Result

Yes, $0.005 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 12$$

Given claim: Mean is less than 6.5

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 6.5$$
$$H_a : \mu < 6.5$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{matrix}0+4+3+4+9+4\\+5+9+1+6+7+10\end{matrix}}{12} = \frac{62}{12} \approx 5.1667$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(0-5.1667)^2 .... + (10-5.1667)^2}{12-1}} \approx 3.1575$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -2.718$$

The rejection region then contains all values smaller than -2.718 and thus we reject H_0 when the t-value is smaller than -2.718.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{5.1667 - 6.5}{3.1575/\sqrt{12}} \approx -1.463$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -1.463 > -2.718 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean number of returns is less than 6.5.

Result

No

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 17$$

Given claim: Mean is less than 3.50

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 3.50$$
$$H_a : \mu < 3.50$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{2.98 + 2.4 + 2.7 + ... + 4.45 + 2.45}{17} = \frac{50.24}{17} \approx 2.9553$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(2.98 - 2.9553)^2 + ... + (2.45 - 2.9553)^2}{17 - 1}} \approx 0.5596$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 17 - 1 = 16$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.746$$

The rejection region then contains all values smaller than -1.746 and thus we reject H_0 when the t-value is smaller than -1.746.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{2.9553 - 3.50}{0.5596/\sqrt{17}} \approx -4.013$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -4.013 < -1.746 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean time for a game is less than 3.50 hours.

Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 28$$
$$\overline{x} = \text{Sample mean} = 42$$
$$s = \text{Sample standard deviation} = 2.1$$

Given claim: Mean is 0

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu = 0$$
$$H_a : \mu \neq 0$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{-0.38 - 0.20 - 0.38 + ... - 0.68 + 0.05}{18} = \frac{-4.15}{18} = -0.2306$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(-0.38 - (-0.2306))^2 + ... + (0.05 - (-0.2306))^2}{18 - 1}} \approx 0.3151$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 18 - 1 = 17$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.110$$

The rejection region then contains all values smaller than -2.110 and all values greater than 2.110,thus we reject H_0 when $t < -2.110$ or $t > 2.110$.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{-0.2306 - 0}{0.3151/\sqrt{18}} \approx -3.105$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -3.105 < -2.110 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim that the mean gain or loss in time for the watches is 0.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 3.105 in the row $df = n - 1 = 18 - 1 = 17$ (two-tailed):

$$0.001 < P < 0.01$$

Note: When using technology, then you obtain the P-value 0.00644.

🎯 **Result**

Reasonable, $0.001 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 365

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 12$$

Given claim: Mean is more than 4.50

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 4.50$$
$$H_a : \mu > 4.50$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{4.63 + 4.15 + 4.76 + ... + 4.52}{12} = \frac{54.86}{12} \approx 4.5717$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(4.63 - 4.5717)^2 + ... + (4.52 - 4.5717)^2}{12 - 1}} \approx 0.2405$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.05$.

$$t = 1.796$$

The rejection region then contains all values greater than 1.796 and thus we reject H_0 when the t-value is larger than 1.796.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{4.5717 - 4.50}{0.2405/\sqrt{12}} \approx 1.033$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 1.033 < 1.796 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean rate of return is more than 4.50 percent.

Result

Not reasonable

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 15$$

Given claim: Mean is more than 100

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 100$$
$$H_a : \mu > 100$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{120 + 108 + 120 + ... + 108 + 117}{15} = \frac{1641}{15} = 109.4$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(120 - 109.4)^2 + ... + (117 - 109.4)^2}{15 - 1}} \approx 9.9628$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value t_0 that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 15 - 1 = 14$ and in the column with $\alpha = 0.05$.

$$t = 2.145$$

The rejection region then contains all values greater than 2.145 and thus we reject H_0 when the t-value is larger than 2.145.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{109.4 - 100}{9.9628/\sqrt{15}} \approx 1.033$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 1.033 < 2.145 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean number of customers using the self-checkout system is more than 100 per day.

🎯 Result

Not reasonable

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 13$$

Given claim: Mean exceeds 267 (increased)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 267$$
$$H_a : \mu > 267$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{321 + 286 + 290 + ... + 281}{13} = \frac{22.4588}{13} = 288.3077$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(321 - 288.3077)^2 + ... + (281 - 288.3077)^2}{13 - 1}} \approx 22.4588$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 13 - 1 = 12$ and in the column with $\alpha = 0.01$.

$$t = 3.055$$

The rejection region then contains all values greater than 3.055 and thus we reject H_0 when the t-value is larger than 3.055.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{288.3077 - 267}{22.4588/\sqrt{13}} \approx 3.421$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 3.421 > 3.055 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean fare has increased.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 3.421 in the row $df = n - 1 = 12 - 1 = 12$ (one-tailed):

$$0.005 < P < 0.01$$

Note: When using technology, then you obtain the P-value 0.00506.

🎯 **Result**

Yes, $0.005 < P < 0.01$

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 10$$
$$\overline{x} = \text{Sample mean} = 1.3$$
$$s = \text{Sample standard deviation} = 0.9$$

Given claim: Mean is 1.5

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu = 1.5$$
$$H_a : \mu \neq 1.5$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value t_0 that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 10 - 1 = 9$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 3.690$$

The rejection region then contains all values smaller than -3.690 and all values greater than 3.690,thus we reject H_0 when $t < -3.690$ or $t > 3.690$.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{1.3 - 1.5}{0.9/\sqrt{10}} \approx -0.703$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$-3.690 < -0.703 < 3.690 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to reject the publisher's claim.

🎯 Result

No

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 25$$
$$\overline{x} = \text{Sample mean} = 30000$$
$$s = \text{Sample standard deviation} = 10000$$

Given claim: Mean is 27000

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu = 27000$$
$$H_a : \mu \neq 27000$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value t_0 that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 25 - 1 = 24$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 3.091$$

The rejection region then contains all values smaller than -3.091 and all values greater than 3.091,thus we reject H_0 when $t < -3.091$ or $t > 3.091$.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{30000 - 27000}{10000/\sqrt{25}} \approx 1.500$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$-3.091 < 1.500 < 3.091 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to reject the claim in the United Nations report.

🎯 Result

No

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$n = \text{Sample size} = 43$$
$$x = \text{Number of successes} = 29$$
$$\alpha = \text{Significance level} = 0.01$$

(a) z is used as the test statistic when we are interested in testing a claim about the population mean when the population standard deviation is known or when we are interested in testing a claim about the population proportion.

In this case, we are interested in testing the claim that the population proportion is 0.5 (as it is believed that each side has the same chance) and thus we need to use z as test statistic.

Step 2

(b) Given claim: p is 0.5 (each side has same chance)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi = 50\% = 0.5$$
$$H_1 : \pi \neq 0.5$$

Step 3

(c) If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Two-tailed test

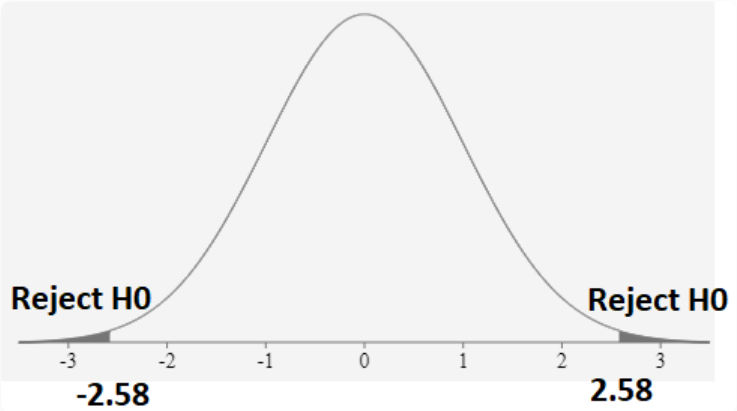
The rejection region of a two-tailed test with $\alpha = 0.01$ contains all z-values below the z-value $-z_0$ that has a probability of $0.01/2 = 0.005$ to its left and all z-values above the z-value z_0 that has a probability of $0.01/2 = 0.005$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.01}{2} = 0.005$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.005 = 0.495$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.495 is closest to 0.4951, which is given in the row starting with "2.5" and in the column starting with ".08", thus the corresponding z-score is $2.5 + .08 = 2.58$.

$$z_0 = \pm 2.58$$

Thus the rejection region then contains all z-scores smaller than -2.58 and all z-scores larger than 2.58, thus we reject H_0 when the z-score is smaller than -2.58 or the z-score is larger than 2.58.



Step 4

(d) The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{29}{43} \approx 0.6744$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.6744 - 0.5}{\sqrt{\frac{0.5(1-0.5)}{43}}} \approx 2.29$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$-2.58 < 2.29 < 2.58 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to reject the claim that the coin flip is fair.

Step 5

(e) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -2.29$ or larger than $z = 2.29$.

$$P(z < -2.29 \text{ or } z > 2.29)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 2.29)$ is given in the row starting with "2.2" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z < -2.29 \text{ or } z > 2.29) \\ &= 2P(z > 2.29) \\ &= 2(1 - P(z < 2.29)) \\ &= 2(1 - P(z < 0) - P(0 < z < 2.29)) \\ &= 2(1 - 0.5000 - 0.4890) \\ &= 2(0.0110) \\ &= 0.0220 \\ &= 2.20\% \end{aligned}$$

🎯 **Result**

(a) Testing a claim about the population proportion

(b) $H_0 : \pi = 0.5, H_1 : \pi \neq 0.5$

(c) Reject H_0 when $z < -2.58$ or $z > 2.58$

(d) There is not sufficient evidence to reject the claim that the coin flip is fair.

(e) $P = 0.0220 = 2.20\%$

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$n = \text{Sample size} = 300$$
$$x = \text{Number of successes} = 210$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is 63%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi = 63\% = 0.63$$
$$H_1 : \pi \neq 0.63$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value z_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than -1.96 and all z-scores larger than 1.96, thus we reject H_0 when the z-score is smaller than -1.96 or the z-score is larger than 1.96.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{210}{300} = 0.7$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.7 - 0.63}{\sqrt{\frac{0.63(1-0.63)}{300}}} \approx 2.51$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$-2.51 < -1.96 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim of the Pet Food Dealers Association.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$n = \text{Sample size} = 200$$
$$x = \text{Number of successes} = 140$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: p is more than 60%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 60\% = 0.60$$
$$H_1 : \pi > 0.60$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all z-scores above the z-score z_0 that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than 2.33 and thus we reject H_0 when the z-score is greater than 2.33.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{140}{200} = 0.7$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.7 - 0.60}{\sqrt{\frac{0.60(1-0.60)}{200}}} \approx 2.89$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$2.89 > 2.33 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that more than 60 percent of the accounts are in arrears for more than three months.

Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$n = \text{Sample size} = 70$$
$$x = \text{Number of successes} = 42$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is more than 55%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 55\% = 0.55$$
$$H_1 : \pi > 0.55$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{42}{70} = 0.6$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.6 - 0.55}{\sqrt{\frac{0.55(1-0.55)}{70}}} \approx 0.84$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$0.84 < 1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the Bowman-to-downtown route meet the STA criterion.

🎯 Result

No

PROBLEM STATEMENT

Chapter 10, Page 366

Step 1

Given:

$$n = \text{Sample size} = 1000$$
$$x = \text{Number of successes} = 480$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is greater than 44% (shift upward)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 44\% = 0.44$$
$$H_1 : \pi > 0.44$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{480}{1000} = 0.480$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.480 - 0.44}{\sqrt{\frac{0.44(1-0.44)}{1000}}} \approx 2.55$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$2.55 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there has been a significant shift upward in the percentage of person who want to go to Europe.

Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$n = \text{Sample size} = 60$$
$$x = \text{Number of successes} = 3$$
$$\alpha = \text{Significance level} = 0.05$$

(a) z is used as the test statistic when we are interested in testing a claim about the population mean when the population standard deviation is known or when we are interested in testing a claim about the population proportion.

In this case, we are interested in testing the claim that the population proportion is 0.10 or 10%, thus we need to use z as test statistic.

Step 2

(b) Given claim: p is 10%\

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi = 10\% = 0.1$$
$$H_1 : \pi \neq 0.1$$

Step 3

(c) If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value z_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than -1.96 and all z-scores larger than 1.96, thus we reject H_0 when the z-score is smaller than -1.96 or the z-score is larger than 1.96.

Step 4

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{3}{60} \approx 0.05$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.05 - 0.10}{\sqrt{\frac{0.10(1-0.10)}{60}}} \approx -1.29$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$-1.96 < -1.29 < 1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to reject the claim in the research report.

Step 5

(d) The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.29$ or larger than $z = 1.29$.

$$P(z < -1.29 \text{ or } z > 1.29)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.29)$ is given in the row starting with "1.2" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z < -1.29 \text{ or } z > 1.29) \\ &= 2P(z > 1.29) \\ &= 2(1 - P(z < 1.29)) \\ &= 2(1 - P(z < 0) - P(0 < z < 1.29)) \\ &= 2(1 - 0.5000 - 0.4015) \\ &= 2(0.0985) \\ &= 0.1970 \\ &= 19.70\% \end{aligned}$$

Result

(a) Testing a claim about the population proportion

(b) $H_0 : \pi = 0.1, H_1 : \pi \neq 0.1$

(c) There is not sufficient evidence to reject the claim in the research report.

(d) $0.1970 = 19.70\%$

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$n = \text{Sample size} = 200$$
$$x = \text{Number of successes} = 56$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: p is greater than 20% (larger proportion)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 20\% = 0.20$$
$$H_1 : \pi > 0.20$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all z-scores above the z-score z_0 that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than 2.33 and thus we reject H_0 when the z-score is greater than 2.33.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{56}{200} = 0.28$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.28 - 0.20}{\sqrt{\frac{0.20(1-0.20)}{200}}} \approx 2.83$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$2.83 > 2.33 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that a larger proportion of condominium owners moved in the Dallas area.

Step 3

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -2.83$ or larger than $z = 2.83$.

$$P(z < -2.83 \text{ or } z > 2.83)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 2.83)$ is given in the row starting with "2.8" and in the column starting with ".03" of the table.

$$\begin{aligned} P &= P(z < -2.83 \text{ or } z > 2.83) \\ &= 2P(z > 2.83) \\ &= 2(1 - P(z < 2.83)) \\ &= 2(1 - P(z < 0) - P(0 < z < 2.83)) \\ &= 2(1 - 0.5000 - 0.4977) \\ &= 2(0.0023) \\ &= 0.0046 \\ &= 0.46\% \end{aligned}$$

🎯 **Result**

Yes, $0.0046 = 0.46\%$

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 8$$

Given claim: Mean is less than 10 (thousand)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 10$$
$$H_a : \mu < 10$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{\begin{array}{c}9.7 + 9.4 + 11.7 + 9.0 \\ +9.1 + 10.5 + 9.1 + 9.8\end{array}}{8} = \frac{20.22}{8} = 9.7875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(9.7 - 9.7875)^2 + ...2 + (9.8 - 9.7875)^2}{8 - 1}} \approx 0.9172$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score z_0 that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = -1.96$$

Thus the rejection region then contains all z-scores smaller than -1.96 and thus we reject H_0 when the z-score is smaller than -1.96.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{9.7875 - 10}{0.9172/\sqrt{8}} \approx -0.655$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$-1.96 < -0.655 < 1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean wedding cost is less than $\$10,000$ as advertised.

Result

Not reasonable

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$n = \text{Sample size} = 30$$
$$x = \text{Number of successes} = 16$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: p is greater than 40% (more likely to skip breakfast)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 40\% = 0.40$$
$$H_1 : \pi > 0.40$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all z-scores above the z-score z_0 that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than 2.33 and thus we reject H_0 when the z-score is greater than 2.33.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{16}{30} \approx 0.5333$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.5333 - 0.40}{\sqrt{\frac{0.40(1-0.40)}{30}}} \approx 1.49$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$1.49 < 2.33 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that college students are more likely to skip breakfast.

🎯 Result

There is not sufficient evidence to support the claim that college students are more likely to skip breakfast.

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$n = \text{Sample size} = 200$$
$$x = \text{Number of successes} = 80$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is less than 50%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \geq 50\% = 0.50$$
$$H_1 : \pi < 0.50$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score z_0 that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.10$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4501, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than -1.65 and thus we reject H_0 when the z-score is smaller than -1.65.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{80}{200} = 0.4$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.4 - 0.50}{\sqrt{\frac{0.50(1-0.50)}{200}}} \approx -2.83$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -2.83 < -1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the proportion of alumni who support the coach is less than 50 percent.

Result

There is sufficient evidence to support the claim that the proportion of alumni who support the coach is less than 50 percent.

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$n = \text{Sample size} = 10000$$
$$x = \text{Number of successes} = 6$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is less than 80 per 100,000 (reduced fatality rate)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \geq \frac{80}{100,000} = 0.0008$$
$$H_1 : \pi < 0.0008$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score z_0 that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4501, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than -1.65 and thus we reject H_0 when the z-score is smaller than -1.65.

Step 2

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{6}{10,000} = 0.0006$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.0006 - 0.0008}{\sqrt{\frac{0.0008(1-0.0008)}{10,000}}} \approx -0.71$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -0.71 > -1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a reduced fatality rate for lung cancer.

🎯 Result

There is not sufficient evidence to support the claim that there is a reduced fatality rate for lung cancer.

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$\alpha = \text{Significance level} = 0.10$$
$$n = \text{Sample size} = 36$$
$$\overline{x} = \text{Sample mean} = 64$$
$$s = \text{Sample standard deviation} = 8.8$$

Given claim: Mean is less than 69

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 69$$
$$H_a : \mu < 69$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed test}$$

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all z-scores below the z-score z_0 that has a probability of 0.10 to its left.

$$P(z < z_0) = 0.10$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.10 = 0.40$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.40 is closest to 0.3997, which is given in the row starting with "1.2" and in the column starting with ".08", thus the corresponding z-score is $1.2 + .08 = 1.28$.

$$z_0 = -1.28$$

Thus the rejection region then contains all z-scores smaller than -1.28 and thus we reject H_0 when the z-score is smaller than -1.28.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{64 - 69}{8.8/\sqrt{36}} \approx -3.409$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -3.409 < -1.28 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that residents of Legacy Ranch use less water on average.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$\alpha = \text{Significance level} = 5\% + 5\% = 10\% = 0.10$$
$$n = \text{Sample size} = 36$$
$$\mu = \text{Population mean} = 9.00$$
$$\sigma = \text{Population standard deviation} = 1.00$$

(a) 5% of the sample means need to be greater than the upper control limit and 5% of the sample means need to be smaller than the lower control limit.

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability closest to 0.45 is 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The lower and upper control limit are z_0 standard deviations from the mean:

$$\mu + z_0 \frac{\sigma}{\sqrt{n}} = 9.00 + 1.65 \frac{1.00}{\sqrt{36}} = 9.275$$
$$\mu - z_0 \frac{\sigma}{\sqrt{n}} = 9.00 - 1.65 \frac{1.00}{\sqrt{36}} = 8.725$$

Thus the lower control limit is 8.725 ounces, while the upper control limit is 9.275 ounces.

Step 2

(b) By part (a), the lower control limit is 8.725 ounces, while the upper control limit is 9.275 ounces.

$$\mu = \text{Population mean} = 8.9$$

The sampling distribution of the sample mean has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{8.725 - 8.9}{1.00/\sqrt{36}} \approx -1.05$$
$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{9.275 - 8.9}{1.00/\sqrt{36}} \approx 2.25$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.05)$ is given in the row starting with "1.0" and in the column starting with ".05" of the table. The probability $P(0 < z < 2.25)$ is given in the row starting with "2.2" and in the column starting with ".05" of the table.

$$\begin{aligned}
P &= P(z < -1.05 \text{ or } z > 2.25) \\
&= P(z < -1.05) + P(z > 1.83) \\
&= P(z > 1.05) + P(z > 1.83) \\
&= 1 - P(z < 1.05) + 1 - P(z < 1.83) \\
&= 1 - P(z < 0) - P(0 < z < 1.05) + 1 - P(z < 0) - P(0 < z < 1.83) \\
&= 1 - 0.5000 - 0.3531 + 1 - 0.5000 - 0.4664 \\
&= 0.1469 + 0.0336 \\
&= 0.1805 \\
&= 18.05\%
\end{aligned}$$

Note: The solution in the back of the book appears to be incorrect, because the solution only takes ir

Step 3

(c) By part (a), the lower control limit is 8.725 ounces, while the upper control limit is 9.275 ounces.

$$\mu = \text{Population mean} = 9.3$$

The sampling distribution of the sample mean has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{8.725 - 9.3}{1.00/\sqrt{36}} \approx -3.45$$
$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{9.275 - 9.3}{1.00/\sqrt{36}} \approx -0.15$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 3.45)$ is given in the row starting with "3.4" and in the column starting with ".05" of the table. The probability $P(0 < z < 0.15)$ is given in the row starting with "0.1" and in the column starting with ".05" of the table.

$$\begin{aligned} P &= P(z < -3.45 \text{ or } z > -0.15) \\ &= P(z < -3.45) + P(z > -0.15) \\ &= P(z > 3.45) + P(z < 0.15) \\ &= 1 - P(z < 3.45) + P(z < 0.15) \\ &= 1 - P(z < 0) - P(0 < z < 3.45) + P(z < 0) + P(0 < z < 0.15) \\ &= 1 - 0.5000 - 0.4997 + 0.5000 + 0.0596 \\ &= 0.0003 + 0.5596 \\ &= 0.5599 \\ &= 55.99\% \end{aligned}$$

Note: The solution in the back of the book appears to be incorrect, because the solution only takes ir

Result

(a) 8.725 ounces and 9.275 ounces

(b) $0.1805 = 18.05\%$

(c) $0.5599 = 55.99\%$

PROBLEM STATEMENT

Chapter 10, Page 367

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 45$$
$$\overline{x} = \text{Sample mean} = 0.80$$
$$\sigma = \text{Population standard deviation} = 0.10$$

(a) Given claim: Mean increased from 0.75

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis needs to include the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \mu \leq 0.75$$
$$H_a : \mu > 0.75$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Step 2

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{0.80 - 0.75}{0.10/\sqrt{45}} \approx 3.35$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 3.35 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean time spent in the mall has increased.

Step 3

(b) By part (a), the null hypothesis is rejected when the z-score is greater than 1.65.

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The control limit is then the mean increased by the product of the z-score and the standard deviation.

$$\text{Control limit} = \mu + z\frac{\sigma}{\sqrt{n}} = 0.75 + 1.65\frac{0.10}{\sqrt{45}} \approx 0.7746$$

Next, we assume that the population mean changed to 0.77.

$$\mu = \text{Population mean} = 0.77$$

The sampling distribution of the sample mean has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-value is the sample mean decreased by the population mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{0.7746 - 0.77}{0.10/\sqrt{45}} \approx 0.31$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 0.31)$ is given in the row starting with "0.3" and in the column starting with ".01" of the table.

$$\begin{aligned} P &= P(z < 0.31) \\ &= P(z < 0) + P(0 < z < 0.31) \\ &= 0.5000 + 0.1217 \\ &= 0.6217 \\ &= 62.17\% \end{aligned}$$

Step 4

(c) Not detecting an increase in the population mean is called a type II error.

The probability of a type II error can be decreased by:

Increasing the sample size (because having more information about the population will allow us to make better estimations).

Increase the significance level (because this increases the probability of making a Type I error and decreases the probability of making a Type II error).

Making the alternative proportion p more extreme (thus p greater than 0.77, since more extreme alternatives are easier to prove).

Result

(a) There is sufficient evidence to support the claim that the mean time spent in the mall has increased.

(b) $0.6217 = 62.17\%$

(c) Increase sample size, Increase significance level

PROBLEM STATEMENT

Chapter 10, Page 368

Step 1

Given:

$$H_0 : \mu \leq 50$$
$$H_1 : \mu > 50$$
$$\alpha = \text{Probability Type I error} = 0.01$$
$$\beta = \text{Probability Type II error} = 0.30$$
$$\mu_a = \text{Alternative mean} = 55$$
$$\sigma = \text{Population standard deviation} = 10$$

Probability type I error

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all z-scores above the z-score z_0 that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than 2.33 and thus we reject H_0 when the z-score is greater than 2.33.

The sampling distribution of the sample mean has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The corresponding control limit is then the mean increased by the product of the z-score and the standard deviation:

$$\overline{x} = \mu + z_0 \frac{\sigma}{\sqrt{n}} = 50 + 2.33 \frac{10}{\sqrt{n}}$$

Step 2

Probability type II error

Type II error: Fail to reject the null hypothesis H_0, when the null hypothesis H_0 is false.

Thus we need to determine the z-score that has a probability of 0.30 to its left.

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.30 = 0.20$ in the normal probability table of the appendix (we need to take the negative of this z-score to obtain the z-score with a probability of 0.30 to its left). We note that the probability 0.20 is closest to 0.1985, which is given in the row starting with "0.5" and in the column starting with ".02", thus the corresponding z-score is $0.5 + .02 = 0.52$.

$$z_0 = -0.52$$

The sampling distribution of the sample mean has mean μ_a and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The corresponding control limit is then the alternative mean increased by the product of the z-score and the standard deviation:

$$\overline{x} = \mu_a + z_0 \frac{\sigma}{\sqrt{n}} = 55 - 0.52 \frac{10}{\sqrt{n}}$$

Step 3

Determine sample size

The two control limits (of the type I error and of the type II error) need to be identical:

$$50 + 1.65\frac{10}{\sqrt{n}} = 55 - 0.52\frac{10}{\sqrt{n}}$$

Multiply each side of the equation by $\sqrt{n}$:

$$50\sqrt{n} + 2.33(10) = 55\sqrt{10} - 0.52(10)$$

Subtract $55\sqrt{n}$ from each side of the previous equation:

$$-5\sqrt{n} + 23.3 = -5.2$$

Subtract 16.5 from each side of the previous equation:

$$-5\sqrt{n} = -28.5$$

Divide each side of the previous equation by -5:

$$\sqrt{n} = 5.7$$

Square each side of the previous equation:

$$n = 5.7^2 = 32.49 \approx 32$$

Thus the sample size is approximately 32.

Result

$$n = 32$$

PROBLEM STATEMENT

Chapter 10, Page 368

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 200$$
$$\overline{x} = \text{Sample mean} = 4800$$
$$s = \text{Sample standard deviation} = 1300$$

Given claim: Mean is less than 5000 (reducing)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 5000$$
$$H_a : \mu < 5000$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed test}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 200 - 1 = 199$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table). *Note: The table does not contain a row with $df = 199$ and thus we use the nearest smaller degrees of freedom $df = 180$ instead.*

$$t = -1.653$$

The rejection region then contains all values smaller than -1.653 and thus we reject H_0 when the t-value is smaller than -1.653.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{4800 - 5000}{1300/\sqrt{200}} \approx -2.176$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -2.176 < -1.653 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the prevention plans were effective in reducing the mean amount of a claim.

🎯 Result

Yes

PROBLEM STATEMENT

Chapter 10, Page 368

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 24$$
$$\overline{x} = \text{Sample mean} = 7.5$$
$$s = \text{Sample standard deviation} = 3.2$$

Given claim: Mean is less than 8

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 8$$
$$H_a : \mu < 8$$

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed}$$

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n - 1 = 24 - 1 = 23$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -2.069$$

The rejection region then contains all values smaller than -2.069 and thus we reject H_0 when the t-value is smaller than -2.069.

Step 2

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{7.5 - 8}{3.2/\sqrt{24}} \approx -0.765$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -0.765 > -2.069 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the waiting time at the local Farmer Jack's is less than that reported in the national magazine.

🎯 Result

No

PROBLEM STATEMENT

Chapter 10, Page 368

Step 1

(a) Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 105$$

Given claim: Mean is more than 220 (thousand)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 220$$
$$H_a : \mu > 220$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{220.9 + 294.5 + 244.6 + ... + 177.1}{105} = \frac{23215.8}{105} = 221.1029$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(220.9 - 221.1029)^2 + ... + (177.1 - 221.1029)^2}{105 - 1}} \approx 47.1054$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value t_0 that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 105 - 1 = 104$ and in the column with $\alpha = 0.01$. *Note: The table does not contain a row with $df = 104$ and thus we use the nearest smaller degrees of freedom $df = 100$ instead.*

$$t = 2.626$$

The rejection region then contains all values greater than 2.626 and thus we reject H_0 when the t-value is larger than 2.626.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{221.1029 - 220}{47.1054/\sqrt{105}} \approx 0.240$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = 0.240 < 2.626 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean selling price in the Goodyear, AZ, area is more than $\$220,000$.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 0.240 in the row $df = n - 1 = 105 - 1 = 104$ (one-tailed). *Note: The table does not contain a row with $df = 104$ and thus we use the nearest smaller degrees of freedom $df = 100$ instead.*

$$P > 0.10$$

Note: When using technology, then you obtain the P-value 0.4054.

Step 4

(b) Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 105$$

Given claim: Mean is more than 2100

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 2100$$
$$H_a : \mu > 2100$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{2300 + 2700 + 2300 + ... + 1900}{105} = \frac{233500}{105} = 2223.8095$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(2300 - 2223.8095)^2 + ... + (1900 - 2223.8095)^2}{105 - 1}} \approx 248.6594$$

Step 7

(c) Given:

$$n = \text{Sample size} = 105$$
$$x = \text{Number of successes} = 71$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is more than 60%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 60\% = 0.60$$
$$H_1 : \pi > 0.60$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed test}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Step 8

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{71}{105} \approx 0.6762$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.6762 - 0.60}{\sqrt{\frac{0.60(1-0.60)}{105}}} \approx 1.59$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = 1.59 < 1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that more than 60 percent of the homes sold in the Goodyear, AZ, area had an attached garage.

Step 9

P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is larger than $z = 1.59$.

$$P(z > 1.59)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.59)$ is given in the row starting with "1.5" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z > 1.59) \\ &= 1 - P(z < 1.59) \\ &= 1 - P(z < 0) - P(0 < z < 1.59) \\ &= 1 - 0.5000 - 0.4441 \\ &= 0.0559 \\ &= 5.59\% \end{aligned}$$

Step 10

(d) Given:

$$n = \text{Sample size} = 105$$
$$x = \text{Number of successes} = 67$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: p is more than 60%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \leq 60\% = 0.60$$
$$H_1 : \pi > 0.60$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed test}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score z_0 that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than 1.65 and thus we reject H_0 when the z-score is greater than 1.65.

Result

(a) No, $P > 0.10$

(b) Yes, $P < 0.0005$

(c) No, $P = 0.0559$

(d) No, $P = 0.2119$

PROBLEM STATEMENT

Chapter 10, Page 368

Step 1

(a) Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 30$$

Given claim: Mean is different from 80.0 (million dollars)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu = 80.0$$
$$H_a : \mu \neq 80.0$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{67.1 + 121.8 + 96.1 + ...60.3}{30} = \frac{2655.4}{30} = 88.5133$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(67.1 - 88.5133)^2 + ... + (60.3 - 88.5133)^2}{30 - 1}} \approx 33.9011$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value t_0 that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 30 - 1 = 29$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.045$$

The rejection region then contains all values smaller than -2.045 and all values greater than 2.045, thus we reject H_0 when $t < -2.045$ or $t > 2.045$.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{88.5133 - 80.0}{33.9011/\sqrt{30}} \approx 1.375$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$-2.045 < 1.375 < 2.045 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean salary of the teams was different from $\$80.0$ million.

Step 3

(b) Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 30$$

Given claim: Mean is more than 2 (million)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \leq 2$$
$$H_a : \mu > 2$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{1.91 + 3.06 + 2.28 + ... + 1.82}{30} = \frac{23215.8}{30} = 2.4477$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(1.91 - 221.1029)^2 + ... + (1.82 - 221.1029)^2}{30 - 1}} \approx 0.6983$$

Result

(a) There is not sufficient evidence to support the claim that the mean salary of the teams was different from $\$80.0$ million.

(b) There is sufficient evidence to support the claim that the mean attendance was more than 2,000,000 per team.

PROBLEM STATEMENT

Chapter 10, Page 368

Step 1

(a) Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 80$$

Given claim: Mean is 840

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu = 840$$
$$H_a : \mu \neq 840$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{853 + 883 + 822 + ... + 449}{80} = \frac{62719}{80} = 783.9875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(853 - 783.9875)^2 + ... + (449 - 783.9875)^2}{80 - 1}} \approx 137.8850$$

Step 2

If the alternative hypothesis H_1 contains $<$, then the test is left-tailed.

If the alternative hypothesis H_1 contains $>$, then the test is right-tailed.

If the alternative hypothesis H_1 contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value t_0 that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 80 - 1 = 79$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 2.640$$

The rejection region then contains all values smaller than -2.640 and all values greater than 2.640, thus we reject H_0 when $t < -2.640$ or $t > 2.640$.

Determine the value of the test statistic:

$$t = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{783.9875 - 840}{137.8850/\sqrt{80}} \approx -3.633$$

We reject the null hypothesis H_0 when the value of the test statistic is in the rejection region.

$$t = -3.633 < -2.640 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean number of miles traveled is equal to 840.

Step 3

P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value 3.633 in the row $df = n - 1 = 80 - 1 = 79$ (two-tailed).

$$P < 0.001$$

Note: When using technology, then you obtain the P-value 0.0005.

Step 4

(b) Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 80$$

Given claim: Mean is less than 500

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu \geq 500$$
$$H_a : \mu < 500$$

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{329 + 503 + 505 + ... + 449}{80} = \frac{36023}{80} = 450.2875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(329 - 450.2875)^2 + ... + (449 - 450.2875)^2}{80 - 1}} \approx 53.6860$$

Step 7

(c) Given:

$$n = \text{Sample size} = 80$$
$$x = \text{Number of successes} = 28$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: p is less than 40%

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis states that the population proportion is equal to the value mentioned in the claim. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi \geq 40\% = 0.40$$
$$H_1 : \pi < 0.40$$

If the alternative hypothesis H_a contains $<$, then the test is left-tailed.

If the alternative hypothesis H_a contains $>$, then the test is right-tailed.

If the alternative hypothesis H_a contains $\neq$, then the test is two-tailed.

$$\text{Left-tailed test}$$

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all z-scores below the z-score z_0 that has a probability of 0.01 to its left.

$$P(z < z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix (the z-score z_0 is then the negative of this z-score). We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = -2.33$$

Thus the rejection region then contains all z-scores smaller than -2.33 and thus we reject H_0 when the z-score is smaller than -2.33.

Step 8

The sample proportion is the number of successes divided by the sample size:

$$\hat{p} = \frac{x}{n} = \frac{28}{80} = 0.35$$

Determine the value of the test-statistic:

$$z = \frac{\hat{p} - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.35 - 0.4}{\sqrt{\frac{0.4(1-0.4)}{80}}} \approx -0.91$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis H_0:

$$z = -0.91 > -1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that less than 40 percent of the buses are old.

Step 9

P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic z, assuming that the null hypothesis is true.

Since the test is left-tailed, the P-value is the probability that the z-score is smaller than $z = -0.91$.

$$P(z < -0.91)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 0.91)$ is given in the row starting with "0.9" and in the column starting with ".01" of the table.

$$\begin{aligned} P &= P(z < -0.91) \\ &= P(z > 0.91) \\ &= 1 - P(z < 0.91) \\ &= 1 - P(z < 0) - P(0 < z < 0.91) \\ &= 1 - 0.5000 - 0.3186 \\ &= 0.1814 \\ &= 18.14\% \end{aligned}$$

🎯 **Result**

(a) Yes, $P < 0.001$

(b) Yes, $P < 0.0005$

(c) No, $P = 0.1814$